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MIL-STD-361-7(TM)
15 JULY 1992

SUPERSEDING
(See 6.5.)

MILITARY STANDARD

MANUALS, TECHNICAL:

OPERATOR AND PREVENTIVE MAINTENANCE INSTRUCTIONS



AMSC A6756

AREA TMSS

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FOREWORD

1. This military standard is approved for use by the Department of the Army is available for use by all Departments and Agencies of the Department of Defense (DOD).

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, USAMC Materiel Readiness Support Activity, ATTN: AMXMD-EP, Lexington, Kentucky 40511-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the document or by letter.

3. This document is one part of an 11-part bookform standard to be used in the preparation and acquisition of technical manuals that are essential to support (operate and maintain) the various types of equipment and weapon systems within the Department of the Army.

4. This document contains operator and preventive maintenance instructions requirements for the preparation of technical manuals.

5. Other parts of this multipart standard are:

MIL-STD-361-1	-	Writing Style and Format Requirements.
MIL-STD-361-2	-	Comprehensibility Requirements.
MIL-STD-361-3	-	Requirements for Graphics.
MIL-STD-361-4	-	Quality Assurance Requirements.
MIL-STD-361-5	-	Requirements for Packaging.
MIL-STD-361-6	-	Introductory Information with Theory of Operations Requirements.
MIL-STD-361-8	-	Maintenance Instructions.
MIL-STD-361-9	-	Troubleshooting Procedures.
MIL-STD-361-10	-	Lubrication Instructions.
MIL-STD-361-11	-	Repair Parts and Special Tools List (RPSTL).

6. By incorporating standardized and service-unique technical manual requirements, this document and its dash-numbered companion parts, along with the general acquisition specification MIL-M-63008, Acquisition/Preparation Requirements, support the acquisition of technical manuals.

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CONTENTS

PARAGRAPH			PAGE
1.	SCOPE		1
	1.1	Scope	1
	1.2	Purpose	1
	1.3	Applicability	1
	1.4	Selective application and tailoring	1
2.	APPLICABLE DOCUMENTS		2
	2.1	Government documents	2
	2.1.1	Specifications, standards, and handbooks	2
	2.1.2	Other Government documents and publications	3
	2.2	Order of precedence	3
3.	DEFINITIONS		4
	3.1	Acronyms used in this standard	4
	3.2	Additional authorization list (AAL) items	5
	3.3	After operation checks	5
	3.4	Auxiliary equipment	5
	3.5	Basic issue items (BII)	5
	3.6	Battlefield damage assessment and repair (BDAR)	5
	3.7	Before operation checks	5
	3.8	Caution	5
	3.9	Component	5
	3.10	Components of end item (COEI)	6
	3.11	Contracting activity	6
	3.12	Disassemble	6
	3.13	During operation checks	6
	3.14	End item	6
	3.15	Equipment	6
	3.16	Equipment, complex	6
	3.17	Fully mission capable (FMC)	6
	3.18	Hardtime scheduled maintenance	6
	3.19	Monthly operation checks	6
	3.20	Note	6
	3.21	Not fully mission capable (NFMC)	6
	3.22	On-condition scheduled maintenance	7
	3.23	Operator maintenance	7
	3.24	Part number (P/N)	7
	3.25	Precombat checklist (PCL)	7
	3.26	Preventive maintenance (scheduled maintenance)	7
	3.27	Replace	7
	3.28	Service	7
	3.29	Spare part	7
	3.30	System	7
	3.31	System integration (SI) manual	7
	3.32	Technical manuals (TMs)	7
	3.33	Unit	8
	3.34	Warning	8
	3.35	Weekly operation checks	8

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Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
4.	GENERAL REQUIREMENTS	9
4.1	General style requirements	9
4.1.1	Conformance to standard	9
4.1.2	Technical content preparation	9
4.1.3	Security classification markings	9
4.1.4	Use of color	9
4.2	Contents	9
4.2.1	Development of text and graphics	9
4.2.2	Operation under unusual conditions	10
4.2.3	Nomenclature and references to equipment	10
4.2.4	Reference to parts of the equipment	10
4.2.5	Abbreviations	10
4.2.6	Illustrations	10
4.2.7	Measurements and tolerances	10
4.2.8	Safety devices and interlocks	10
4.2.9	Glossary	10
4.2.10	Advertising	10
5.	DETAILED REQUIREMENTS	11
5.1	General	11
5.1.1	Operating procedures in multilevel maintenance TMs	11
5.2	Preparation of TMs	11
5.2.1	Text content	11
5.2.2	Troubleshooting procedures	11
5.2.3	Operating instructions	11
5.3	Arrangement of operator's TM (-10)	11
5.3.1	Chapters, sections, appendixes, index	12
5.3.2	Front matter	12
5.4	Chapter 1 - introduction	12
5.5	Chapter 2 - operating instructions	12
5.5.1	Section I - description and use of operator's controls and indicators	12
5.5.2	Section II - preventive maintenance checks and services (PMCS) and mandatory replacement parts	13
5.5.2.1	Introduction to PMCS table	13
5.5.2.2	Equipment illustration	14
5.5.2.3	PMCS table	14
5.5.2.3.1	Mandatory replacement parts	16
5.5.2.4	Preventive maintenance checklist (PMC)	16
5.5.2.4.1	Illustrations	16
5.5.2.4.2	Style and format requirements	16
5.5.2.4.3	Technical content	17
5.5.2.4.4	Warnings and cautions	17
5.5.2.4.5	Usage note and reporting errors statement	17
5.5.2.5	Precombat checklist (PCL)	18
5.5.2.5.1	Illustrations	18
5.5.2.5.2	Style and format requirements	18
5.5.2.5.3	Technical content	18
5.5.2.5.3.1	Cover sheet	18
5.5.2.5.3.2	Checklist	19

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:31Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
5.5.3	Section III - operation under usual conditions	20
5.5.3.1	Site and shelter requirements	20
5.5.3.1.1	Siting	20
5.5.3.1.2	Shelter requirements	21
5.5.3.2	Assembly and preparation for use	21
5.5.3.3	Initial adjustments, before use, daily checks, and self-test (as applicable)	21
5.5.3.4	Operating procedure	22
5.5.3.4.1	Method of presentation	22
5.5.3.5	Decals and instruction plates	23
5.5.3.6	Operating auxiliary equipment	23
5.5.3.7	Preparation for movement	23
5.5.4	Section IV - operation under unusual conditions	23
5.5.4.1	Unusual environment/weather	24
5.5.4.2	Fording and swimming	24
5.5.4.3	Emergency procedures	24
5.5.4.4	Nuclear, biological, and chemical (NBC) decontamination procedures	24
5.5.4.5	Jamming and electronic countermeasures (ECM) procedures	24
5.5.5	Section V - functional checks	24
5.6	Chapter 3 - operator maintenance instructions	24
5.6.1	Section I - lubrication instructions	24
5.6.2	Section II - troubleshooting procedures	25
5.6.2.1	Arrangement of procedures	25
5.6.2.2	Introductory information	25
5.6.2.3	Malfunction index	25
5.6.3	Section III - operator maintenance procedures	26
5.7	Chapter 4 - auxiliary equipment maintenance	26
5.8	Chapter 5 - ammunition/class V material	26
5.9	Chapter 6 - foreign ammunition (NATO)	26
5.10	Appendixes	26
5.10.1	References appendix	26
5.10.2	Components of end item (COEI) and basic issue items (BII) lists appendix	26
5.10.2.1	Section I - introduction	27
5.10.2.2	Section II - components of end item	28
5.10.2.3	Section III - basic issue items	28
5.10.2.4	Stowage location	28
5.10.3	Additional authorization list (AAL) appendix	28
5.10.3.1	Section I - introduction	29
5.10.3.2	Section II - additional authorized items list	29
5.10.4	Expendable and durable items list appendix	29
5.10.5	Stowage and decal/data plate guide appendix	30
5.10.6	On-vehicle equipment loading plan appendix	30
5.10.6.1	Illustrations	31

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T07:31Z
Check the source to verify that this is the current version before use.

CONTENTS

PARAGRAPH		PAGE
5.10.6.2	Listing	31
5.10.7	Lubrication instructions appendix	31
5.10.8	Other appendixes	31
5.10.9	Glossary	31
5.10.10	Alphabetical index	31
5.11	Quality assurance requirements	31
5.12	Packaging and preparation for delivery	31
6.	NOTES	32
6.1	Intended use	32
6.2	Issue of DODISS	32
6.3	Tailoring guidance	32
6.4	Acquisition requirements	32
6.5	Supersession data	32
6.6	Subject term (key word) listing	33

FIGURES		PAGE
1.	Example of controls and indicators for equipment text format (reference 5.5.1a)	34
2.	Example of a PMCS introduction (sheet 1 of 3) (reference 5.5.2.1)	35
2.	Example of a PMCS introduction (sheet 2 of 3) (reference 5.5.2.1)	36
2.	Example of a PMCS introduction (sheet 3 of 3) (reference 5.5.2.1)	37
3.	Example of operator's PMCS table format (reference 5.5.2.3)	38
4.	Example of operator's PMCS table format with crewmember grouping (reference 5.5.2.3c(2))	39
5.	Example of PMCS mandatory replacement parts list (reference 5.5.2.3.1)	40
6.	Example of a PMC cover (reference 5.5.2.4.2, 5.5.2.4.5)	41
7.	Example of an accordion foldout (reference 5.5.2.4.2c)	42
8.	Example of a PMC inside page identifying crewmembers (reference 5.5.2.4.3d)	43
9.	Example of PCL cover page (reference 5.5.2.5.2b, 5.5.2.5.3.1)	44
10.	Example of a checklist for a TM with one PMCS table (reference 5.5.2.5.3.2a)	45
11.	Example of a checklist for a TM with more than one PMCS table (reference 5.5.2.5.3.2a)	46
12.	Example of a checklist for a TM with more than one volume (reference 5.5.2.5.3.2a)	47
13.	Example of operation procedure format (sheet 1 of 2) (reference 5.5.3)	48
13.	Example of operation procedure format (sheet 2 of 2) (reference 5.5.3)	49
14.	Example of NBC presentation (sheet 1 of 2) (reference 5.5.4.4)	50

Source: https://assist.dla.mil -- Downloaded: 202610-26T07:31Z
Check the source to verify that this is the current version before use.

CONTENTS

FIGURES PAGE

14. Example of NBC presentation (sheet 2 of 2) (reference 5.5.4.4) 51
15. Example of a typical troubleshooting table (reference 5.6.2.1) 52
16. Example of a troubleshooting malfunction index (reference 5.6.2.3) 53
17. Example of an ammunition identification illustration (reference 5.8) 54
18. Example of a typical ammunition table (sheet 1 of 3) (reference 5.8) 55
18. Example of a typical ammunition table (sheet 2 of 3) (reference 5.8) 56
18. Example of a typical ammunition table (sheet 3 of 3) (reference 5.8) 57
19. Example of references appendix (reference 5.10.1) . . 58
20. Example of a components of end item list (reference 5.10.2.2) 59
21. Example of a basic issue items list (reference5.10.2.3, 5.10.2.4) 60
22. Example of format for additional authorized items (reference 5.10.3.2) 61
23. Example of an expendable and durable items list format (reference 5.10.4) 62
24. Example of stowage and decal/data plate guide appendix (reference 5.10.5) 63
25. Example of a tactical on-vehicle loading plan (reference 5.10.6) 64
26. Example of a non-tactical on-vehicle loading plan (reference 5.10.6) 65

INDEX 66

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1. SCOPE

1.1 Scope. This document establishes the content and procedural requirements for preparing operator instructions and preventive maintenance instructions for technical manuals (TMs), revisions, supplements, and changes.

1.2 Purpose. This document supplements the various content requirements in the other parts to this 11-part standard for specific types of technical manuals and related publications and does not alone deliver any technical data. (NOTE: The figures used in this document are examples only. The text of this document takes precedence over the figures.)

1.3 Applicability. The applicability of the requirements contained in this document is governed by the equipment, systems, or weapon systems for which the technical manuals are being developed. This document is applicable for use by the Department of the Army.

1.4 Selective application and tailoring. This document contains some requirements that may not be applicable to the preparation of all technical manuals. Selective application and tailoring of requirements contained in this document are the responsibility of the contracting activity and shall be accomplished through the use of MIL-M-63008. The applicability of some requirements is also designated by one of the following statements:

- unless requirement is specifically excluded by the contracting activity; or
- as/when specified by the contracting activity.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation. (See 6.2.)

SPECIFICATIONS

MILITARY

MIL-M-63008	-	Manuals, Technical: Acquisition/ Preparation Requirements.

STANDARDS

MILITARY

MIL-STD-12	-	Abbreviations for Use on Drawings, and in Specifications, Standards, and Technical Documents.
MIL-STD-361-1	-	Manuals, Technical: Writing Style and Format Requirements.
MIL-STD-361-2	-	Manuals, Technical: Comprehensibility Requirements.
MIL-STD-361-3	-	Manuals, Technical: Requirements for Graphics.
MIL-STD-361-4	-	Manuals, Technical: Quality Assurance Requirements.
MIL-STD-361-5	-	Manuals Technical: Requirements for Packaging.
MIL-STD-361-6	-	Manuals, Technical: Introductory Information with Theory of Operations Requirements.
MIL-STD-361-8	-	Manuals, Technical: Maintenance Instructions.
MIL-STD-361-9	-	Manuals, Technical: Troubleshooting Procedures.
MIL-STD-361-10	-	Manuals, Technical: Lubrication Instructions.

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MIL-STD-1388-2 - DOD Requirements for a Logistic Support Analysis Record.

HANDBOOKS

MILITARY

MIL-HDBK-491A - Grounding, Bonding, and Shielding for Electronic Equipment and Facilities.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5904.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

AMC-P 750-2 - Guide to Reliability Centered Maintenance.

DOD 5320.24 - Distribution Statements on Technical Documents.

(Copies of other Government documents and publications required by the contractor in connection with specific acquisition functions should be obtained from the contracting activity. Copies of DOD Directive 5320.24 can be obtained from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document ("This document" includes all volumes of MIL-STD-361 series.) and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms used in this standard. The acronyms used in this standard are defined follows:

a. AAL - Additional Authorization List.

b. BDAR - Battlefield Damage Assessment and Repair.

c. BII - Basic Issue Items.

d. BIT - Built-In Test.

e. CAGEC - Commercial and Government Entity Code.

f. COEI - Components of End Item.

g. DOD - Department of Defense.

h. DODISS - Department of Defense Index of Specifications and Standards.

i. ECM - Electronic Countermeasures.

j. FMC - Fully Mission Capable.

k. LOAP - List of Applicable Publications.

l. LSA - Logistic Support Analysis.

m. LSAR - Logistic Support Analysis Record.

n. MAC - Maintenance Allocation Chart.

o. MNS - Mission Needs Statement.

p. NBC - Nuclear, Biological, Chemical.

q. NFMC - Not Fully Mission Capable.

r. NSN - National Stock Number.

s. PCL - Precombat Checklist.

t. PMC - Preventive Maintenance Checklist.

u. PMCS - Preventive Maintenance Checks and Services.

v. RCM - Reliability Centered Maintenance.

w. RGL - Reading Grade Level.

x. ROC - Requirements for Operational Capability.

y. RPSTL - Repair Parts and Special Tools List.

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z. SI - System Integration.

aa. SMR - Source, Maintenance, Recoverability.

ab. TM - Technical Manual.

ac. UOC - Usable On Code.

3.2 Additional authorization list (AAL) items. Items not issued with the end item, not listed on the end item engineering drawing as part of the end item configuration, not required to be turned in with the end item, separately authorized, and provided for information only.

3.3 After operation checks. After operation checks are those checks that are performed immediately at the conclusion of the mission and that identify and correct faults that will prevent the next mission, ensure the equipment is fully mission capable (FMC), and maintain the equipment to an original condition standard. Faults which render the equipment not fully mission capable (NFMC) must be corrected prior to start of the next mission. All other faults are corrected or, if above operator level, reported to unit maintenance and action taken before next mission.

3.4 Auxiliary equipment. Equipment, accessories, or devices that, when used with basic equipment, extend or increase its capability (e.g., Modified Table of Organization and Equipment (MTOE) items, etc.).

3.5 Basic issue items (BII). Essential items required to place the equipment in operation, operate it, and to do emergency repairs. These items are shipped separately packaged, but must be with the equipment during operation and when it is transferred between property accounts.

3.6 Battlefield damage assessment and repair (BDAR). Procedure to rapidly return disabled equipment to the operational commander by expediently fixing, bypassing, or jury rigging components to restore the minimum essential systems required for the support of a specific combat mission or to enable the equipment to self-recover.

3.7 Before operation checks. Before operation checks are those checks that are performed just prior to the start of the mission and that identify faults which will prevent the performance of the mission.

3.8 Caution. A statement or some other notification about an operating or maintenance procedure, practice, or condition that, if not strictly observed, could result in damage to, or destruction of, equipment or loss of mission effectiveness.

3.9 Component. A constituent part not normally considered to be capable of independent operation; a piece part.

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3.10 Components of end item (COEI). Spare/repair parts that are removed from the major end item and separately packaged or stowed for transportation or movement; includes on-board spares.

3.11 Contracting activity. The Department of Defense (DOD) component, activity, or organization of a using military service, or that organization delegated by a using service, that is responsible for the selection and determination of requirements for TMs.

3.12 Disassemble. The step-by-step taking apart (or breakdown) of a spare or functional group-coded item to the level of its least componency identified as maintenance-significant (i.e., assigned a source, maintenance, and recoverability (SMR) code) for the category of maintenance under consideration.

3.13 During operation checks. During operation checks are those checks that are performed during the mission and that identify faults in equipment performance. Faults which render the equipment NFMC require immediate correction. These checks should be very minimal and observable while operating the equipment.

3.14 End item. A final combination of end products, component parts, or materials that is ready for its intended use (e.g., tank, mobile machine shop, aircraft, receiver, rifle, recorder).

3.15 Equipment. One or more units capable of performing specified functions.

3.16 Equipment, complex. Equipment with numerous sophisticated systems installed, including integrated systems and built-in test (BIT) capability.

3.17 Fully mission capable (FMC). Equipment that can perform all of its combat missions without further endangering the lives of crew or operators in a combat environment.

3.18 Hardtime scheduled maintenance. Hardtime maintenance is scheduled maintenance conducted at predetermined fixed intervals because of age, calendar, or usage such as operating time, flying hours, miles driven, or rounds fired.

3.19 Monthly operation checks. Monthly operation checks are those checks that are performed for faults which do not need to be checked for weekly but must be checked for more often than at next service by unit maintenance and which must be corrected to sustain equipment to FMC standards until the next service by unit maintenance.

3.20 Note. A statement or some other notification that adds, emphasizes, or clarifies essential information of special importance or interest.

3.21 Not fully mission capable (NFMC). Equipment that cannot perform all of its combat missions without further endangering the lives of the crew in a combat environment.

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3.22 On-condition scheduled maintenance. On-condition scheduled maintenance is maintenance or item replacement action performed based upon condition of the item as determined by an evaluation of each item on a scheduled basis.

3.23 Operator maintenance. Operator maintenance consists of inspecting, servicing, lubricating, adjusting, replacing, and repairing those items authorized by logistic support analysis (LSA) and/or maintenance allocation chart (MAC).

3.24 Part number (P/N). A primary number used to identify an item used by the manufacturer (individual, company, firm, corporation, or Government activity) that controls the design, characteristics, and production of the item by means of its engineering drawings, specifications, and inspection requirements.

3.25 Precombat checklist (PCL). Checks to be performed before combat and during alerts to assure combat vehicle/equipment is FMC.

3.26 Preventive maintenance (scheduled maintenance). The performance of scheduled inspections and maintenance functions necessary to keep the equipment in serviceable condition and ready for its primary mission.

3.27 Replace. To remove an unserviceable spare or repair part and install a serviceable counterpart in its place. Replace is authorized by the LSA/MAC and the assigned maintenance level is shown as the third position code of the SMR code.

3.28 Service. Operations required periodically to keep an item operating, i.e., to clean (includes decontaminate, when required), to preserve, to drain, to paint, or to replenish fuel, lubricants, chemical fluids, or gases.

3.29 Spare part. Any reparable and recoverable component required for the maintenance or repair of an end item or equipment (will have a recoverability code other than "Z" and will be assigned an FGC or subfunctional group code in the MAC).

3.30 System. A regularly interacting or interdependent group of items forming a unified whole to perform a specific operational function.

3.31 System integration (SI) manual. A manual that is used when a system is composed of multiple components with each component having its own TM coverage. It is used to provide the system overview and as the starting point for all operation troubleshooting and maintenance procedures.

3.32 Technical manuals (TMs). Manuals that contain instructions for the installation, operation, maintenance, and support of weapon systems, weapon system components, and support equipment. TM information may be presented, according to prior agreement between the contractor and the Government, in any form or

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characteristic, including hard printed copy, audio and visual displays, electronic imbedded media, disks, other electronic devices, or other media. They normally include operational and maintenance instructions, parts lists, and related technical information or procedures exclusive of administrative procedures.

3.33 Unit. An assembly or any combination of spare and repair parts mounted together, normally capable of independent operation in a variety of situations.

3.34 Warning. A statement or some other notification about an operating or maintenance procedure, practice, or condition that, if not strictly observed, could result in long term health hazard, injury to, or death of personnel performing the task prescribed in the TM.

3.35 Weekly operation checks. Weekly operation checks are those checks that are performed to identify faults which must be corrected to sustain equipment to FMC standards until next service by unit maintenance.

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4. GENERAL REQUIREMENTS

4.1 General style requirements.

4.1.1 Conformance to standard. Preparation of cover page, front matter, text, security classification, classification markings, numbering, changes, warnings, cautions, health hazards, notes, final reproducible copy of final manual, and other document elements shall be in accordance with MIL-STD-361-1.

4.1.2 Technical content preparation. The following baseline requirements shall be used, as specified by the contracting activity:

a. Logistic support analysis record (LSAR). The technical data and instructions developed by the requirements of MIL-STD-1388-2, DOD Requirements for a Logistic Support Analysis Record (LSAR), shall be used as the baseline to prepare TMs.

b. Maintenance allocation chart (MAC). The MAC shall be used as the baseline to prepare TMs. The MAC shall be prepared from the equipment top-down breakdown and be in functional group code sequence to consolidate and identify those groups on the list which involve identified maintenance functions. Logistic Support Analysis (LSA) Output Report 04, Maintenance Allocation Summary will be used when available as source data for the final approved MAC.

When specified by the contracting activity and if a waiver of standard requirements is granted, TB 750-93-1 may be used as the baseline for preparing TMs.

4.1.3 Security classification markings. All classified TMs and material shall be safeguarded and identified in accordance with MIL-STD-361-1.

4.1.4 Use of color. Use of color shall be avoided. If approved by the contracting activity, color shall be used in accordance with MIL-STD-361-1 and MIL-STD-361-3.

4.2 Contents. The TM shall describe in accurate and appropriate detail the operation and maintenance authorized for the operator/crew. Sufficient information shall be included in procedures and illustrations so that personnel can operate and maintain the equipment safely and efficiently. The TM shall be written for the user in accordance with the target audience description and Reading Grade Level (RGL) provided by the contracting activity. Unless requirement is specifically excluded by the contracting activity, the TM will include identification of specific crew members and their assigned tasks.

4.2.1 Development of text and graphics. The requirements of MIL-STD-361-1, MIL-STD-361-2, and MIL-STD-361-3 shall be applicable to the development of text and graphics. Reference to

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information contained elsewhere shall be in accordance with MIL-STD-361-1. Procedural steps shall be worded and arranged in the order that will provide the most effective and efficient results. The expected results shall be presented and the corrective measures to be taken shall be provided when such results are not obtained.

4.2.2 Operation under unusual conditions. The TM shall include information on any special operator and preventive maintenance procedures required under extreme temperature, altitude, humidity, or other environmental conditions within the limits established by the design specification for the equipment. These procedures shall give due consideration to the fact that the tactical situation may dictate system operation under conditions exceeding the design limits of the system.

4.2.3 Nomenclature and references to equipment. Nomenclature and references to equipment shall be in accordance with the requirements of MIL-STD-361-1.

4.2.4 Reference to parts of the equipment. Reference to parts of the equipment and to equipment components shall be consistent between the repair parts and special tools list (RPSTL), LSA/MAC, and maintenance TMs. Reference shall be by type designation, standard nomenclature, reference designation, or figure and index number, as applicable.

4.2.5 Abbreviations. Abbreviations shall be in accordance with MIL-STD-361-1 and MIL-STD-12.

4.2.6 Illustrations. Illustrations shall be in accordance with MIL-STD-361-2 and MIL-STD-361-3. Only illustrations and information that apply to the operator shall be used.

4.2.7 Measurements and tolerances. All numerical data shall be presented in a consistent, clear, easily understood manner. (Refer to MIL-STD-361-1, MIL-STD-361-2, and MIL-STD-361-3.)

4.2.8 Safety devices and interlocks. The purpose and location of all safety devices and interlocks shall be given in conjunction with the pertinent procedures.

4.2.9 Glossary. Glossaries shall be used in TMs only when the terms are uncommon and are not adequately defined in the text or in the Army, DOD, or standard dictionary. When required, a glossary shall be prepared in accordance with MIL-STD-361-1.

4.2.10 Advertising. Except for the identity of the equipment manufacturer or contractor, publication material prepared in accordance with this standard shall contain no advertising matter.

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5. DETAILED REQUIREMENTS

5.1 General. These requirements shall apply to operator TMs for equipment and systems within the Department of the Army. As specified by the contracting activity, either separate TMs shall be prepared for operation and for preventive maintenance, or a combined operator and preventive maintenance TM shall be prepared. Writing style and format shall be in accordance with MIL-STD-361-1 and MIL-STD-361-3.

5.1.1 Operating procedures in multilevel maintenance TMs. Operating instructions shall be combined with maintenance TMs in accordance with MIL-M-63008 and MIL-STD-361-8, when specified by the contracting activity. The following titles are furnished as examples of typical combined operator and maintenance TMs:

Operator and Unit Maintenance Manual (-12)

Operator and Aviation Unit Maintenance Manual (-12)

Operator, Unit, and Direct Support Maintenance Manual (-13)

Operator, Aviation Unit, and Aviation Intermediate Maintenance Manual (-13)

Operator, Unit, Direct Support, and General Support Maintenance Manual (-14)

5.2 Preparation of TMs.

5.2.1 Text content. Collectively, the TMs shall contain all essential information required to operate, maintain, troubleshoot, and support the equipment for which these TMs are prepared.

5.2.2 Troubleshooting procedures. TMs prepared in accordance with this standard shall describe in appropriate detail the troubleshooting procedures at the operator/crew level identified by the LSA/MAC, 5.6.2, and MIL-STD-361-9.

5.2.3 Operating instructions. The operating instructions shall describe:

a. How, in general, the item performs its function.

b. How to prepare the item for initial use.

c. How to operate the item under both ordinary and emergency conditions.

5.3 Arrangement of operator's TM (-10). The operator's TM shall describe in detail the operator level of maintenance allocated by the LSA/MAC. The arrangement of the TM shall depend on the equipment the TM will cover. The following list contains the basic requirements:

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Front Matter
Chapter 1. Introduction
Chapter 2. Operating Instructions
Chapter 3. Operator Maintenance Instructions
Chapter 4. Auxiliary Equipment Maintenance Instructions
Chapter 5. Ammunition/Class V Material
Chapter 6. Foreign Ammunition (NATO)

APPENDIXES
References
Components of End Item (COEI) and Basic Issue Items (BII)
Additional Authorization List (AAL)
Expendable and Durable Items List
Stowage and Decal/Data Plate Guide
On-Vehicle Equipment Loading Plan
Lubrication Instructions
Glossary
Alphabetical Index

5.3.1 Chapters, sections, appendixes, index. Unless requirement is specifically excluded by the contracting activity, if a chapter, section, or appendix does not apply to a particular item of equipment, then the remaining chapters, sections, or appendixes shall be renumbered so that no gaps appear in the numbering sequence.

5.3.2 Front matter. Front matter shall be prepared in accordance with MIL-STD-361-6 (content) and MIL-STD-361-1 (format).

5.4 Chapter 1 - introduction. This chapter shall be prepared in accordance with MIL-STD-361-6.

5.5 Chapter 2 - operating instructions. This chapter shall present instructions needed by the operator to operate the equipment. The level of detail shall be appropriate for the user. This chapter shall consist of the following sections:

Section I. Description and use of operator's controls and indicators

Section II. Preventive maintenance checks and services (PMCS) and mandatory replacement parts

Section III. Operation under usual conditions

Section IV. Operation under unusual conditions

Section V. Functional checks (for SI manuals)

5.5.1 Section I - description and use of operator's controls and indicators.

a. This section shall include line drawings of all operator controls and indicators. Each control and indicator shall be clearly labeled as it appears on the equipment.

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Controls and indicators that are not labeled, such as the accelerator or brake pedals, shall be identified. The functional use of each control and indicator shall be explained by use of a table(s), keyed text, or listing that describes each control separately. See figure 1 for example.

b. For system integration (SI) manuals, Section I shall be titled "Summary of Operations." This section shall present an overview of the system. This overview shall describe how the end items that compose the system interrelate to achieve system functionality. When necessary, line drawings shall be used to support the text and to show signal/data flow, equipment relationships, and functional block diagrams for system integration. When the system has some system level controls and indicators that are not covered by other TMs, the description and use of operator's controls and indicators shall be included for these controls. For systems/equipment not requiring separate SI manuals, this overview shall be part of Chapter 1, Section III, Theory of Operation.

5.5.2 Section II - preventive maintenance checks and services (PMCS) and mandatory replacement parts. This section shall contain a PMCS table with checks and services that are based on reliability centered maintenance (RCM), as prescribed in AMC-P 750-2. The table shall contain only those checks and services authorized to the operator by the MAC under "inspection" and "service" functions. For SI manuals, the PMCS table shall begin with system level checks (if applicable), then proceed to reference the end item TMs for their PMCS, and shall conclude with referencing the paragraph for the system functional check. The table shall have an introduction which explains to the user its purpose and how to use it. When a change for a combat vehicle is made to the PMCS table that changes an item number of the current procedure, a change to the combat vehicle precombat checklist (PCL) is required.

5.5.2.1 Introduction to PMCS table. The first paragraph in Section II shall be Introduction to PMCS Table. The introduction shall contain the following subparagraphs:

a. General. This paragraph shall contain an explanation of the purpose and use of the PMCS table.

b. PMCS procedures. This paragraph shall contain an explanation of each column of the table and any general checks/services that are common to the entire piece of equipment.

c. Special information paragraphs. These paragraphs shall contain any special information such as cleaning agents, lubrication, and fluid leakage information that is necessary for the user to know before doing PMCS.

For an example of a PMCS introduction, see figure 2.

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5.5.2.2 Equipment illustration. Unless requirement is specifically excluded by the contracting activity, the first page after the introduction shall include an illustration of the equipment. The illustration shall include a routing diagram showing the order by which the checks and services will be done.

5.5.2.3 PMCS table. The PMCS table shall be formatted in accordance with the example in figure 3 and shall include the following:

a. Special instructions.

(1) For equipment that is normally kept in continuous operation, the following note shall be included between the title of the table and the table.

"NOTE
If the equipment must be kept in continuous operation, do only the procedures that can be done without disturbing operation. Make complete checks and services when the equipment is shut down."

(2) When check and service intervals must be shortened because the equipment may be used under unusual conditions, an asterisk (*) shall precede the interval. A footnote shall explain the asterisk and the reason for the shortened interval.

(3) A statement such as the following shall be included, when applicable:

When a check and service procedure is required for both "weekly" and "before" intervals, it is not necessary to perform the "weekly" procedure during the same week in which the "before" procedure was done.

(4) When the equipment contains fluids, such as lubrication oil or hydraulic fluid, leakage criteria shall be included in the introduction to the PMCS and referred to in the NOT MISSION CAPABLE IF column.

b. Title. The table shall have a title as follows:

"Preventive Maintenance Checks and Services for (insert equipment model number)."

c. Arrangement of procedures. PMCS procedures shall be arranged in a logical sequence so that a minimum amount of time and motion is used without crewmembers interfering with each other.

(1) Interval groupings. Check and service procedures shall be grouped according to their intervals. Procedures done first or most frequently, such as

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"before" checks and services, shall appear before "during" and "after" checks and services. Unless requirement is specifically excluded by the contracting activity, intervals beyond weekly shall not be used in the operator's PMCS table. Intervals determined by operation, such as hours of operation, shall be expressed in the applicable units of measure. Unless requirement is specifically excluded by the contracting activity, reference to other procedures shall not be made. Each procedure shall stand alone.

(2) Crewmember groupings. When more advantageous to the user, intervals shall be subgrouped by crewmember(s) as shown on figure 4.

d. Components of end item (COEI) and basic issue items (BII). All COEI and BII that require individual PMCS procedures shall be separately incorporated into the end item PMCS table under their applicable group/title headings.

e. Warnings, cautions, and notes. Warnings, cautions, and notes shall appear in the PMCS table as prescribed by MIL-STD-361-1.

f. PMCS table columnar headings.

(1) Item No. column. Unless requirement is specifically excluded by the contracting activity, item numbers shall be assigned to procedures in consecutive numerical order.

(2) Interval column. The interval column shall indicate when the checks and services are to be done. The intervals shall be represented by complete words such as: "Before" for before operation; "During" for during operation; and "After" for after operation. See section 3 of this standard for definitions of these intervals.

(3) Location, item to check/service column. This column shall contain the name of the item(s) to be checked or serviced. When appropriate, the location of the item within the equipment/end item shall be provided, underlined, and appear above the item name. The header shall include the word "Location" above "Item to Check/Service." When the location is given, the introduction to the PMCS table shall explain its use.

(4) Procedure column. This column shall contain concise procedures required to do the checks and services. The procedure shall begin with an action word and shall be stated in the second person imperative mood. The procedure shall include all information needed to do the checks and services, including

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appropriate tolerances, adjustments, limits, and gage and meter readings. The procedure column shall have illustrations as required to identify the location or the process of task being performed. Unless requirement is specifically excluded by the contracting activity, illustrations shall be integrated with the procedures. (See figure 3.) Unless requirement is specifically excluded by the contracting activity, reference to procedures and information in other parts of the document or in other documents shall not be made.

(5) Not fully mission capable if: column. Entries in this column shall be keyed specifically to checks and services appearing in the procedure column. Entries shall identify the faults of the equipment, determined by the checks and services procedures, that will prevent the equipment from being fully mission capable.

5.5.2.3.1 Mandatory replacement parts. This paragraph shall contain a table listing all items that must be replaced during PMCS, whether they have failed or not. The table shall reflect the interval at which these items must be replaced, whether hardtime maintenance or on-condition maintenance is the determining factor. Refer to figure 5.

5.5.2.4 Preventive maintenance checklist (PMC). Unless requirement is specifically excluded by the contracting activity, a PMC shall be prepared. Detailed requirements for the preparation of an operator's pocket size PMC for all equipment except aircraft are provided. The PMC includes all BEFORE, DURING, and AFTER operation preventive maintenance checks, including tactical and safety checks, that the operator or crew performs to ensure that the equipment is fully mission capable.

5.5.2.4.1 Illustrations. Illustrations shall not be used in the PMC.

5.5.2.4.2 Style and format requirements.

a. The general and style and format of the PMC shall be in accordance with MIL-STD-361-1.

b. The PMC shall consist of a cover page (See figure 6.) and the checklist. The checklist shall consist of all BEFORE, DURING, and AFTER operation check and service procedures in the PMCS table in the associated operator's TM.

c. When the content of the PMC, including the cover page, is 10 pages or less, it shall be prepared as a pocket size accordion foldout. An example is shown in figure 7. If it is 11 pages or more, it shall be prepared as a vertical pocket manual. The PMC shall have a trim size of 4 by 5-1\2 inches and a maximum printed area shall include all printed matter including publication and page numbers.

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d. Type size shall not be smaller than 6 point for pocket size accordion foldout and 8 point for the vertical pocket manual format.

e. The PMC shall use the same basic TM identification number as the operator's TM from which the preventive maintenance checks and services were extracted. A -10PMC suffix shall be added to the basic TM number. See figure 6 for an example.

5.5.2.4.3 Technical content.

a. The checklist shall consist of all BEFORE, DURING, and AFTER operation interval check and service procedures contained in the PMCS table in the associated operator's TM.

b. The item numbers in the checklist shall be the same as those assigned to the procedures in the operator's PMCS table.

c. The check and service procedures in the checklist shall be presented in the same sequence and include the same crewmember headings, if any, as those in the associated operator's PMCS table. Appropriate interval headings shall precede each group of procedures.

d. Any check that could reveal a not fully mission capable fault shall be identified by placing an asterisk (*) by that item number. Each page containing an asterisk shall have a note explaining its meaning and instructing the operator to report the not fully mission capable fault using the appropriate forms. The note shall appear at the bottom of the page. See figure 8.

5.5.2.4.4 Warnings and cautions. All warnings and cautions applicable to the check and service procedures extracted from the PMCS table in the appropriate operator's TM shall be included in the checklist in their entirety.

5.5.2.4.5 Usage note and reporting errors statement. The following text shall appear on the front cover of the PMC (figure 6):

"NOTICE

To effectively perform the tasks in this checklist, you must be experienced in using the PMCS table in TM (insert the identification number of the applicable operator's TM). The item numbers in this checklist are the same as those in the PMCS table.

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes, or if you know of a way to improve the procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank

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Forms), direct to: (name and address of proponent). A reply will be furnished to you."

5.5.2.5 Precombat checklist (PCL). Unless requirement is specifically excluded by the contracting activity, a PCL shall be prepared for combat vehicles. The PCL shall include all before, during, after, and weekly operation checks, including tactical and safety, that the operator or crew performs to ensure that the equipment is fully mission capable and ready to shoot, move, and communicate.

5.5.2.5.1 Illustrations. Illustrations shall not be used in the PCL.

5.5.2.5.2 Style and format requirements.

a. The general style and format of the PCL shall be in accordance with MIL-STD-361-1.

b. The PCL shall consist of one sheet, front and back: a cover page (See figure 9.) and the checklist (See figure 10.). The checklist shall consist of before, during, after, and weekly operation checks from the PMCS table necessary to ensure that the equipment can do its designated combat mission.

c. The PCL shall be prepared pocket size. The PCL shall have a trim size of 5 by 8 inches and a maximum printed area of 4-3/4 by 7-3/4 inches. The maximum printed area shall include all printed matter including publication and page number. The checklist of the PCL shall be numbered Arabic numeral 2. Page 1, the cover, shall not be numbered.

d. Type size shall not be smaller than 8 points. The PCL shall be printed on white 110 index cardstock and shall be laminated on both sides with 0.005-inch thick clear polyester with a minimum overall thickness of 0.019 inch. The PCL shall have a 3/8-inch hole punched in the upper left-hand corner, 5/16 inch from hole center to top and side of card.

e. The PCL shall use the same basic TM identification number as the operator's TM from which the PMCS item was extracted. The suffix designation -PCL, instead of -10, shall be added to the basic TM number. The publication number shall be placed in the top center of the cover.

5.5.2.5.3 Technical content.

5.5.2.5.3.1 Cover sheet. The cover sheet shall consist of the following in the order given:

a. Publication number. The publication number of a PCL shall be the same basic TM identification number as the operator's TM with a -PCL in place of -10. The

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publication number shall be placed in the top center of the front cover as shown in figure 9.

b. Title block. The title block of the PCL shall consist of the system/equipment type (e.g., combat vehicle), the words "PRECOMBAT CHECKLIST FOR:", and the name of the system/equipment. See figure 9.

c. Warning and note. The cover sheet shall have the following warning and note verbatim:

"**WARNING**
THIS CHECKLIST IS TO BE USED ONLY WHEN AUTHORIZED BY THE COMMANDER. IT IS NOT INTENDED TO BE USED IN PLACE OF THE PMCS.

NOTE
THIS CHECKLIST CONTAINS CHECKS TO BE PERFORMED BEFORE COMBAT TO ASSURE VEHICLE IS MISSION CAPABLE. THE CHECKLIST SEQUENCE NUMBERS (AND THE TABLE NUMBERS OR TM VOLUME NUMBERS, as applicable) HEREIN CORRESPOND TO THOSE PMCS CHECKS IN (INSERT OPERATOR TM NUMBER). REPORT CHECK RESULTS TO YOUR IMMEDIATE LEADER."

The above warning and note shall be placed on the cover sheet as shown on figure 9.

d. Distribution statement. The distribution statement shall be in accordance with DOD Directive 5320.24.

e. Authentication block. The authentication block shall be added by the Government and shall be positioned on the cover sheet as shown on figure 9.

f. Special distribution instructions. Any special distribution instructions shall be placed on the cover sheet as shown on figure 9.

g. Military service. The proponent military service (e.g., Department of the Army) shall be centered below the special distribution instructions. See figure 9.

h. Publication date. The publication date of the PCL shall be flush right below the military service as shown on figure 9.

5.5.2.5.3.2 Checklist.

a. Format. The checklist shall be formatted depending on what the PMCS table/operator's TM looks like. If there is one PMCS table, the checks shall be listed without a header. See figure 10. If there is more than one PMCS table, the checks shall be listed by table number. See figure 11. If the operator's TM has more than one volume, the checks shall be listed by volume number (e.g., 10-1,

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10-2). See figure 12. The checks shall have the same numbers as the checks in the PMCS table from which they were selected. Each check shall consist of a title (e.g., vehicle exterior) printed in all capital letters and a brief explanation/summary of the check, printed in capital and small letters. See figures 10, 11, and 12 for examples of checks.

b. Criteria for selection of checks. The checks to be included in the combat vehicle PCLs shall be tasks from the operator's PMCS table. These checks shall generally be, but are not limited to, before operation, mission-essential, maintenance/operation checks. The checks shall be in a logical sequence, and the crew members shall be able to perform all checks in the PCL in 20 minutes or less. The checks selected shall be those checks determined to be mission-essential based on the Requirements for Operational Capability (ROC), the Mission Needs Statement (MNS), and/or other requirements (e.g., safety). The PCL shall consist of the minimum essential checks required to be sure fire control, armament, communications, and automotive equipment are functioning and the system/end item can perform its primary mission (i.e., move, shoot, and communicate).

5.5.3 Section III - operation under usual conditions. Step-by-step instructions shall be given for all actions necessary to operate the equipment and auxiliary equipment for all modes of operation. Any combination of control settings that will cause damage to the equipment or create a hazard to personnel shall be preceded by a caution or a warning. Appropriate cautions and warnings shall be included to alert the user to possible damage or injury that can result. Instructions to ensure proper grounding of equipment shall be included in accordance with MIL-HDBK-491A. Figure 13 depicts typical operating procedures. As applicable, the instructions shall cover the areas described in the following paragraphs.

5.5.3.1 Site and shelter requirements.

5.5.3.1.1 Siting.

a. Siting instructions peculiar to the equipment shall be detailed and illustrated. Operational features shall be considered, such as:

(1) Location.

(2) Proximity to power sources.

(3) Effective ranges.

(4) Terrain requirements to avoid screening reflections, ground clutter, and other poor operational conditions due to terrain.

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(5) Technical requirements.

(6) Shelter locations.

b. As applicable, instructions on compensating for adverse siting conditions shall be included.

c. When the equipment being covered contains large components such as towers and antennas that require orientation to a baseline during siting, the orientation procedure shall be covered under siting.

d. If the equipment is mobile and the type that is oriented during installation, the orientation procedure shall be covered.

5.5.3.1.2 Shelter requirements. Whenever the equipment will normally be housed in a permanent or semipermanent shelter (other than a military truck, van, or transportable shelter) during use, the TM shall state:

a. Amount of floor and wall space required.

b. A plan for a typical layout.

c. Required weight capacity of the building floor.

d. Dimensions required for installed equipment.

e. Total weight that the floor must support and the area in square feet over which the total weight will be distributed, if several different types of equipment are installed.

f. Environmental conditions (e.g., venting).

NOTE

Architectural and engineering data on beam sizes, lengths, bending moments, and required supports shall not be included.

5.5.3.2 Assembly and preparation for use. This paragraph shall include unpacking, assembly, and installation instructions. When the equipment is shipped/delivered in specially designed containers, unpacking instructions shall be provided. If the containers are to be used again, kept for future use, turned in to supply, or if any special disposition is required, the necessary procedures shall be included. Step-by-step assembly and installation procedures shall be provided when needed. These instructions shall be supported by illustrations. As applicable, power requirements, connections, and initial control settings needed for installation purposes shall be included.

5.5.3.3 Initial adjustments, before use, daily checks, and self-test (as applicable). Instructions shall be given for any

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routine checks, self-test, or adjustments that the operator must make before putting the equipment in operation.

5.5.3.4 Operating procedure. Instructions shall be given to place the equipment in operation, to operate the equipment in each mode of operation for which it was intended, to remove the equipment from operation and put it in standby condition, and to remove it from operation and put it in shutdown status. These procedures must explain how the equipment is operated in conjunction with auxiliary equipment and show integration with other equipment, as applicable. Operator instructions shall include misfire, hangfire, and other procedures applicable to ammunition. In addition, the following shall be addressed:

a. All steps necessary to bring the equipment from OFF through STANDBY condition to full operation.

b. Procedures for each mode of operation, e.g., manual, automatic, local, remote, etc. The use and relative advantage of each mode shall also be described.

c. Description of the equipment anti-jamming and interference reduction features, the advantage of each feature, and the operating procedures to be followed. Supporting illustrations (such as indicator displays, waveforms, etc.) shall be included which provide typical observations of jamming and interference for evaluation by the operator.

d. Operator turn-off procedures including all steps necessary to bring the equipment from full operation through STANDBY to OFF condition.

e. Procedures covering operation of the equipment during emergency conditions (control failure, air failure, lube oil failure, loss of cooling water, etc.). Emergency operator instructions shall be included. Provide a warning or caution to return the equipment to proper operation when the emergency is over.

f. Procedures to turn the equipment off during an emergency (fire, water, smoke, hazard to personnel, loss of coolant, normal power, etc.).

5.5.3.4.1 Method of presentation. Operating procedures shall be presented in concise, simply-worded, step-by-step procedures and shall include the following, as applicable:

a. Initial safety requirements (actions, inspections, and emergency turn-off procedures).

b. An explanation of the operation to be performed.

c. Connection of any accessory equipment not permanently connected.

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d. Instructions for obtaining or confirming the presence of all critical inputs such as power, coolant, air, signal, air-conditioning, etc.

e. Procedures for setting controls and making adjustments which must be accomplished by the operator prior to equipment turn-on.

f. Procedures for determining operational readiness and the acceptable indications expected from built-in indicators such as meters, lamps, gages, cathode ray tubes, and recorder readouts.

g. Milestones in the operational status of the equipment, indicated by brief statements, such as "the generator is now in STANDBY."

h. Visual or audible observations which occur as a result of an operator action, such as boom lowering, sweep rotation, blower motor running, etc.

i. Procedures that can be hazardous to personnel or damage equipment shall be emphasized by WARNINGs or CAUTIONs placed immediately prior to the specific step involving the possible hazard.

j. Illustrations supporting the procedures shall identify and locate all operating controls and indicating devices as well as normal in-use positions or indications.

5.5.3.5 Decals and instruction plates. Decals and operating instruction plates located on the equipment, essential for operation, shall be clearly illustrated, so that all data are legible. Related warning and caution decals and plates shall be included. An illustration(s) shall be used to show the location of all applicable decals and plates.

5.5.3.6 Operating auxiliary equipment. If applicable, this paragraph shall include procedures to put the auxiliary equipment into operation, to operate it, and to put it in standby or shutdown status. If these procedures are published in another TM covering the auxiliary equipment, reference shall be made to that TM in accordance with the guidelines of MIL-STD-361-1.

5.5.3.7 Preparation for movement. If the equipment is designed for movement, and it can be prepared for movement by the operator, preparation procedures shall be provided. These procedures shall include information for actions such as disassembly, folding, and telescoping. Essential illustrations shall be used to support the text.

5.5.4 Section IV - operation under unusual conditions. This section shall contain instructions for operation under unusual conditions including special operating instructions for each expected condition. These instructions shall include protective and preventive measures to be taken for these conditions.

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Preventive or protective measures to be taken beyond the operator's capabilities shall be identified and referenced or action to be taken by the operator shall be provided. Instructions to ensure proper grounding of equipment shall be included in accordance with MIL-HDBK-491A. The following conditions shall be covered, as applicable.

5.5.4.1 Unusual environment/weather. Instructions shall include procedures for operation under conditions of extreme moist heat, extreme dry heat, extreme cold, salt air, sea spray, dust storms, sand storms, high altitudes, snow, mud, and other similar conditions. Define ranges of environmental/weather operating conditions considered for the system addressed.

5.5.4.2 Fording and swimming. Instructions shall include special procedures required before, during, and after fording and swimming the equipment.

5.5.4.3 Emergency procedures. Instructions shall be included for temporarily adapting the equipment and the operating procedure to meet the reduction of power, partial failure, failure of a portion of the equipment, or similar conditions when continued equipment use is required.

5.5.4.4 Nuclear, biological, and chemical (NBC) decontamination procedures. As applicable and specified by the contracting activity, interim general NBC decontamination procedures to performed until NBC decontamination facilities are available shall be provided. Other decontamination documents shall be referenced only when necessary. For an example of an interim general decontamination procedures presentation, see figure 14.

5.5.4.5 Jamming and electronic countermeasures (ECM) procedures. As applicable, instructions shall be included for operation of the equipment in an ECM environment through transmitted and reflected deception signals and transmitted and reflected jamming.

5.5.5 Section V - functional checks. For SI manuals, and when specified by the contracting activity for non-SI manuals, this section shall contain all operational checks that must be performed to confirm that the system/equipment is fully operational.

5.6 Chapter 3 - operator maintenance instructions. This chapter shall include sections covering lubrication, troubleshooting, and maintenance procedures.

5.6.1 Section I - lubrication instructions. When equipment lubrication instructions are published, reference shall be made to the appropriate document. When lubrication instructions are not published, this section shall contain one of the following statements, as applicable:

a. "Lubrication Not Required."

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b. "Lubrication instructions are in appendix _, of this TM. All lubrication instructions are mandatory."

5.6.2 Section II - troubleshooting procedures. This section shall contain information to help the operator recognize, find the cause, and correct the trouble in the equipment and auxiliary equipment. For systems with SI manuals, troubleshooting for any fault shall begin in the SI manual. Troubleshooting shall continue in the SI manual until the fault or symptom can be isolated to a specific component. At this point, further troubleshooting shall be referenced to the system component TM in such a manner that the correct troubleshooting path in the system component TM is easily recognized. The information shall be presented by simplified illustrations, charts, tables, or diagrams in the format most suitable for the equipment. Refer also to MIL-STD-361-9.

5.6.2.1 Arrangement of procedures. The procedures shall be arranged so that the most probable malfunctions (trouble) appear first. Each procedure, or set of procedures, shall assume the following logic: the malfunction shall be stated first, followed by test or inspection procedure(s) (presented in logical order to isolate and identify the fault), and action required to correct the fault. When corrective action is extensive and detailed instructions appear in Section III, reference shall be made to the paragraph(s) covering that information. When the equipment contains several major components, the troubleshooting procedures shall be grouped by major components. For example of a troubleshooting table, see figure 15.

5.6.2.2 Introductory information. Tables, charts, and diagrams shall have an introduction. It shall give a brief explanation of the presentation, how to use it, and include statements similar to the following:

a. This table lists all the common malfunctions that you may find with your equipment. Perform the tests, inspections, and corrective actions in the order they appear in the table.

b. This table cannot list all the malfunctions that may occur, all the tests and inspections needed to find the fault or all the correction actions needed to correct the fault. If the equipment malfunction is not listed or actions listed do not correct the fault, notify your supervisor.

5.6.2.3 Malfunction index. Unless requirement is specifically excluded by the contracting activity, a malfunction index shall precede the table. The index shall list the malfunctions that may be observed by the operator. They shall be listed in the same order that they appear in the table. The malfunction listed shall be cross-referenced to the paragraph or location where troubleshooting information can be found. See figure 16.

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5.6.3 Section III - operator maintenance procedures. This section shall contain procedures for maintenance functions authorized to the operator by the LSA/MAC. All procedures shall be preceded with an introduction. Step-by-step procedures shall be presented in the order in which they should be done and supported by illustrations. Procedures shall include functions such as inspect, test, service, install, replace, disassemble and assemble, repair, clean, adjust, and align.

5.7 Chapter 4 - auxiliary equipment maintenance. This chapter shall be included in the TM when there is auxiliary equipment. The chapter shall contain all maintenance information applicable to the auxiliary equipment as required by 5.6.

5.8 Chapter 5 - ammunition/class V material. This chapter shall be included in all operator's TMs on weapons involving the use of ammunition/class V material. Only that information needed by the operator for the weapons shall be included. An illustration shall be used to clearly show all identifying markings on ammunition. See figure 17 for a typical example. A table shall be included covering classification, identification, care, handling, preservation, transportation, authorized rounds, preparation for firing, fuzes, packing, and other pertinent data and references, as appropriate. See figure 18 for examples. Reusable original packaging and containers shall be identified for return or temporary storage of ammunition in its original configuration. Information on classifying, identifying, caring for, handling, etc., nonammunition class V items shall be provided, when applicable.

5.9 Chapter 6 - foreign ammunition (NATO). A chapter to describe foreign ammunition shall be included unless requirement is specifically excluded by the contracting activity. The requirements of 5.8 shall apply to this chapter.

5.10 Appendixes. Include as applicable.

5.10.1 References appendix. This appendix shall list all publications referenced in the TM and required by the operator to operate and maintain the equipment. The appendix shall have an introduction or scope paragraph to provide a brief statement(s) concerning its use and content. The publications shall be listed in groups by publication types. If the publication is non-Government, the source shall be provided. The complete name of each publication and the publication number shall be used. See figure 19. When a list of applicable publications (LOAP) is published, this appendix shall reference the LOAP.

5.10.2 Components of end item (COEI) and basic issue items (BII) lists appendix. This appendix shall consist of two lists: the Components of End Item List and the Basic Issue Items List. Section I of the appendix shall be the introduction to both lists. Section II shall consist of the COEI listing and illustrations of the items, as necessary. Section III shall consist of the BII listing and illustrations, as necessary.

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5.10.2.1 Section I - introduction. Introduction format and content shall be in accordance with the following:

"APPENDIX _

COMPONENTS OF END ITEM AND BASIC ISSUE ITEMS LISTS

SECTION I. INTRODUCTION

_-1 SCOPE.

This appendix lists components of end item and basic issue items for the (insert the short end item name) to help you inventory items for safe and efficient operation of the equipment.

_-2 GENERAL.

The Components of End Item (COEI) and Basic Issue Items (BII) Lists are divided into the following sections:

a. Section II - Components of End Item. This listing is for information purposes only and is not authority to requisition replacements. These items are part of the (enter name of end item). As part of the end item, these items must be with the end item whenever it is issued or transferred between property accounts. Items of COEI are removed and separately packaged for transportation or shipment only when necessary. Illustrations are furnished to help you find and identify the items.

b. Section III - Basic Issue Items. These essential items are required to place the (enter name of end item) in operation, operate it, and to do emergency repairs. Although shipped separately packaged, BII must be with the (enter name of end item) during operation and when it is transferred between property accounts. Listing these items is your authority to request/requisition them for replacement based on authorization of the end item by the TOE/MTOE. Illustrations are furnished to help you find and identify the items.

_-3 EXPLANATION OF COLUMNS.

a. Column (1), Illus Number, gives you the number of the item illustrated.

b. Column (2), National Stock Number, identifies the stock number of the item to be used for requisitioning purposes.

c. Column (3), Description, CAGEC, Part Number, and Used on Code, identifies the Federal item name (in all capital letters) followed by a minimum description when needed. The last line below the description is the CAGEC (commercial and Government entity code) (in parentheses) and the part number. (Add the following only as applicable.) If the item you need is not the same for different models of equipment, a Used On Code will appear on the right side of the description column on the same line as the part number. These codes are identified below:

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CODE	USED ON
XXX	Model XXX
XXX	Model XXXX
XXX	Model XXXXX

d. Column (4), U/M (unit of measure), indicates how the item is issued for the National Stock Number shown in column (2).

e. Column (5), Qty Rqr, indicates the quantity required."

5.10.2.2 Section II - components of end item. This section shall consist of illustrations and a listing of components of the end item (spare/repair parts that are removed from the major end item and separately packaged or stowed for transportation or movement; includes on-board spares). The illustrations shall be line drawings and shall be placed above or precede the listing. The arrangement of the illustrations and listing shall be similar to that shown on figure 20.

a. Illustrations. All items appearing in the listing shall be illustrated. Each item shall be identified with a circled number. Item number one or the lowest item number listed on the page, shall be located at the upper left area of the illustration field. The other items shall be placed in numerical order from left to right or in clockwise order.

b. Listing. The COEI list shall include the headings and basic content shown on figure 20, applicable to the specific equipment. The description of each item shall consist of the approved Federal item name, followed by a short description when needed. Items shall be listed alphabetically. The CAGEC shall be located below the item and in parentheses. The part number shall follow the CAGEC. When more than one model or configuration is involved and Used on codes (UOC) are assigned, the UOC shall appear at the right edge and on the last line of the description column for the applicable component. When on-board spares apply, there shall be a break in the text of the listing and new heading "ON-BOARD SPARES" shall be used. A listing of the on-board spares shall appear in the same format as required for the basic COEI list.

5.10.2.3 Section III - basic issue items. This section shall be prepared in the same format and include the same content (tailored to the applicable BII) as required for the COEI list. See figure 21.

5.10.2.4 Stowage location. The stowage location of COEI and BII shall be included in the description column of the COEI and BII lists. See figure 21 for example.

5.10.3 Additional authorization list (AAL) appendix. This appendix shall list all AAL items (items not issued with the end item, not listed on the end item engineering drawing as part of

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the end item (NSN) configuration, not required to be turned in with the end item, separately authorized by an MTOE, TDA, CTA, or JTA, and provided for information only). The appendix shall consist of two sections: Section I, Introduction, and Section II, Additional Authorized Items List. No illustrations shall be used.

5.10.3.1 Section I - introduction. The introduction shall include the same format and general content as follows:

"APPENDIX _

ADDITIONAL AUTHORIZATION LIST

SECTION I. INTRODUCTION

_-1 SCOPE.

This appendix lists additional items you are authorized for the support of the (enter short item name).

_-2 GENERAL.

This list identifies items that do not have to accompany the (enter short item name) and that do not have to be turned in with it. These items are all authorized to you by CTA, MTOE, TDA, or JTA.

_-3 EXPLANATION OF LISTING.

National stock numbers, description, and quantities are provided to help you identify and request the additional items you require to support this equipment. If the item required differs for different models of this equipment, see the "Used on Code" column for the applicable model or models. Codes used are:

CODE	USED ON
XXX	Model XXX
XXX	Model XXXX
XXX	Model XXXXX"

5.10.3.2 Section II - additional authorized items list. This section shall list additional authorized items. The format and general content of the list shall be prepared as shown on figure 22. The headings and subsequent information for this listing shall be the same as the COEI and BII lists except the Illus Number Column required for the COEI and BII lists shall not apply since there are no illustrations used, and the Qty column shall be Qty Recm (quantity recommended). The items shall be listed alphabetically.

5.10.4 Expendable and durable items list appendix. This appendix shall have two sections. Section I shall be the introduction. Wording shall be adapted to the specific application. Section II shall be the tabular list prepared in

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the format shown in figure 23. No illustrations shall be provided for these items. Items appearing in the tabular list shall appear in alphabetical sequence by item name. Items to be listed shall be those approved by the contracting activity. This list shall include the items authorized to the operator/crew. Following is a representation of the general format and content to be used for preparing Section I, Introduction:

"APPENDIX _

EXPENDABLE AND DURABLE ITEMS LIST

SECTION I. INTRODUCTION

_-1 SCOPE.

This appendix lists expendable and durable items that you will need to operate and maintain the (enter item name). This listing is for information only and is not authority to requisition the listed items. These items are authorized to you by CTA 50-790, Expendable/Durable Items (except medical, class V repair parts, and heraldic items), or CTA 8-100, Army Medical Department Expendable/Durable Items.

_-2 EXPLANATION OF COLUMNS.

a. **Column (1) - Item Number.** This number is assigned to the entry in the listing and is referenced in the narrative instructions to identify the item (e.g., "Use cleaning compound (item 5, Appendix D).").

b. **Column (2) - Level.** This column identifies the lowest level of maintenance that requires the item.

c. **Column (3) - National Stock Number.** This is the National stock number assigned to the item which you can use to requisition it.

d. **Column (4) - Item Name, Description, Commercial and Government Entity Code (CAGEC), and Part Number.** This provides the other information you need to identify the item.

e. **Column (5) - Unit of Measure (U/M).** This code shows the physical measurement or count of an item, such as gallon, dozen, gross, etc."

5.10.5 Stowage and decal/data plate guide appendix. This appendix shall include an illustration detailing the location of applicable COEI, BII, and AAL items. As applicable, this appendix shall also include illustrations detailing the locations of all decals and data plates. See figure 24 for example.

5.10.6 On-vehicle equipment loading plan appendix. This appendix shall be included, unless requirement is specifically excluded by the contracting activity. The loading plan shall include information provided by the contracting activity. The

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first page of the appendix shall identify the equipment covered and shall include a brief scope statement explaining the purpose of the loading plan. The plan shall include illustrations and tabular listings of the items identified on the illustrations. For examples, see figures 25 and 26.

5.10.6.1 Illustrations. Line drawing illustrations shall show the location of all applicable equipment. External and internal views shall be used. As applicable, both tactical and non-tactical situation loading configurations shall be shown. Items shall be identified with numerical callouts and arrowhead leader-lines. The item numbers shall be assigned to follow a logical sequential order, such as clockwise, left to right, or top to bottom.

5.10.6.2 Listing. The listing (legend) shall be located on the same page or the page adjacent to the associated illustration(s). The list shall have two columns, the item number column (corresponding to the illustration item numbers) and the item name.

5.10.7 Lubrication instructions appendix. This appendix shall include all lubrication instructions as specified in MIL-STD-361-10.

5.10.8 Other appendixes. Other appendixes shall be added as applicable, and when approved by the contracting activity.

5.10.9 Glossary. Unless requirement is specifically excluded by the contracting activity, a glossary shall be included and prepared in accordance with MIL-STD-361-1.

5.10.10 Alphabetical index. The TM shall contain an alphabetical subject index, prepared in accordance with MIL-STD-361-1.

5.11 Quality assurance requirements. Quality assurance and quality control for TMs, revisions, supplements, and changes shall be in accordance with MIL-STD-361-4.

5.12 Packaging and preparation for delivery. TMs, revisions, supplements, and changes shall be packaged in accordance with MIL-STD-361-5.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This document prescribes the operator and preventive maintenance requirements applicable to various types of technical manuals, revisions, supplements, and changes prepared by or for the Department of the Army.

6.2 Issue of DODISS. When this document is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation. (See 2.1.1.)

6.3 Tailoring guidance. The contracting activity should tailor any required options offered herein in accordance with MIL-M-63008.

6.4 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this standard.

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced. (See 2.1.1)

c. Target audience description and reading grade level (4.2).

d. Expendable/durable items to be listed (5.10.4).

e. On-vehicle equipment loading information (5.10.6).

6.5 Supersession data. This document consolidates, standardizes, and supersedes the operator and preventive maintenance instructions requirements contained in the following documents:

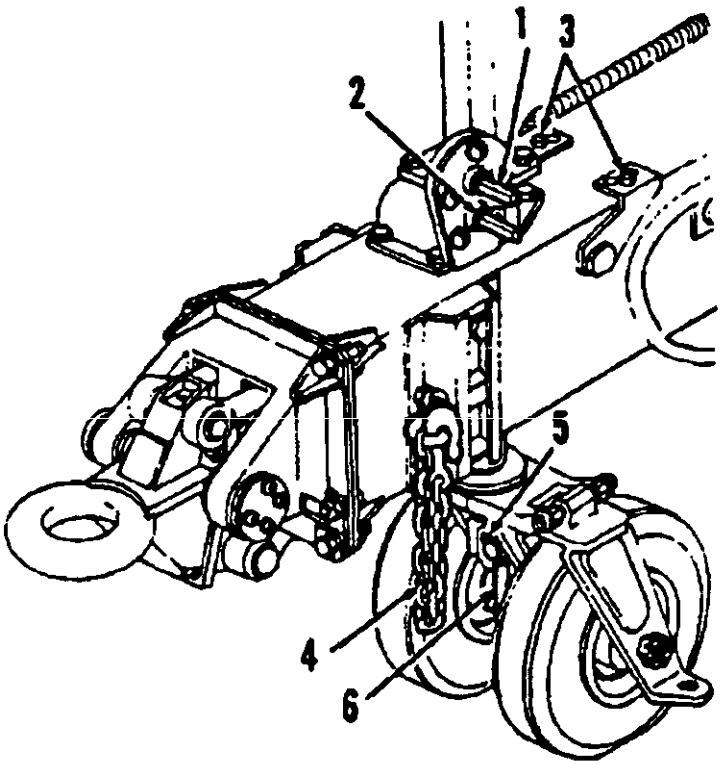
MIL-M-49504(TM)	-	Manuals, Technical: Combat Vehicle Precombat Checklist.
MIL-M-63006(TM)	-	Manuals, Technical: Daily Preventive Maintenance Checklist.
MIL-M-63036D(TM)	-	Manuals, Technical: Operator's, Preparation of.
MIL-M-63038C(TM)	-	Manuals, Technical: Unit or Aviation Unit, Direct Support, Aviation Intermediate, and General Support Maintenance, Requirements for.

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6.6 Subject term (key word) listing. The following terms are to be used to identify this document during retrieval searches:

Additional authorization list (AAL)
Alphabetical index
Ammunition/class V material
Appendixes
Arrangement of operator's manual
Assembly and preparation for use
Basic issue items (BII)
Components of end item (COEI)
Decals and instruction plates
Description and use of operator's controls and indicators
Emergency procedures
Expendable and durable items list
Foreign ammunition (NATO)
Jamming and electronic countermeasures (ECM)
Lubrication instructions
Maintenance instructions
Maintenance of auxiliary equipment
Nuclear, biological, and chemical (NBC) decontamination
On-vehicle equipment loading plan
Operating auxiliary equipment
Operating instructions
Operating procedures in multilevel maintenance manuals
Operation under unusual conditions
Operation under usual conditions
Operator's manual
Precombat checklist (PCL)
Preparation for movement
Preventive maintenance checklist (PCM)
Preventive maintenance checks and services (PMCS)
References
Site and shelter requirements
Stowage and decal/data plate guide
System integration (SI) manual
Troubleshooting procedures

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KEY	CONTROL OR INDICATOR	FUNCTION
1	Cranking stud (High Speed)	Extends or retracts caster jack.
2	Cranking stud (Low Speed)	Extends or retracts caster jack and allows 2-man operation with missile round on board.
3	Caster up lock stud (2)	Engages with hole in lug on caster forks to secure caster wheels in stowed position.
4	Tow bar safety chain (2) shown in stowed position	Provides a safety factor when the tow bar is attached to a towing vehicle.
5	Recess for pry bar	Facilitates steering of caster wheels on difficult terrain.
6	Caster down lockpin	Locks caster wheels in down position.

FIGURE 1. Example of controls and indicators for equipment text format (reference 5.5.1a).

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SECTION II. PREVENTIVE MAINTENANCE CHECKS AND SERVICES (PMCS) AND MANDATORY REPLACEMENT PARTS

2.4 INTRODUCTION TO PMCS TABLE.

2.4.1 General.

Preventive Maintenance Checks and Services (PMCS) means systematic caring, inspecting, and servicing of equipment to keep it in good condition and to prevent breakdowns. As the truck's operator, your mission is to:

a. Be sure to perform your PMCS each time you operate the truck. Always do your PMCS in the same order, so it gets to be a habit. Once you've had some practice, you'll quickly spot anything wrong.

b. Do your BEFORE PMCS just before you operate the truck. Pay attention to WARNINGs, CAUTIONs, and NOTEs.

c. Do your DURING PMCS while you operate the truck. During operation means to monitor the truck and its related components while it is actually being operated. Pay attention to WARNINGs, CAUTIONs, and NOTEs.

d. Do your AFTER PMCS right after operating the truck. Pay attention to WARNINGs, CAUTIONs, and NOTEs.

e. Do your WEEKLY PMCS once a week.

f. Use DA Form 2404 (Equipment Inspection and Maintenance Worksheet) to record any faults that you discover before, during, or after operation, unless you can fix them. You DO NOT need to record faults that you fix.

g. Be prepared to assist unit maintenance when they lubricate the truck. Perform any other services when required by unit maintenance.

2.4.2 PMCS Procedures.

a. Your Preventive Maintenance Checks and Services, Table 2-1, lists inspections and care required to keep your truck in good operating condition. It is set up so you can make your BEFORE OPERATION checks as you walk around the truck.

b. The "ITEM NO." column of Table 2-1 contains a number for each item (sequential numbering). The "INTERVAL" column tells you when to do a certain check or service.

c The "LOCATION/ITEM TO CHECK/SERVICE" column lists the location and name of the item(s).

d. The "PROCEDURE" column of Table 2-1 tells you how to do required checks and services. Carefully follow these instructions. If you do not have tools, or if the procedure tells you to, notify your supervisor.

NOTE

Terms "ready/available" and "mission capable" refer to same status: Equipment is on hand and ready to perform its combat missions. (See DA Pam 738-750.)

e. The "NOT FULLY MISSION CAPABLE IF:" column in Table 2-1 tells you when your truck is nonmission capable and why the truck cannot be used.

f. If the truck does not perform as required, refer to Chapter 3, Section II, Troubleshooting.

g. If anything looks wrong and you can't fix it, write it on your DA Form 2404. IMMEDIATELY, report it to your supervisor.

FIGURE 2. Example of a PMCS introduction (sheet 1 of 3) (reference 5.5.2.1).

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h. When you do your PMCS, you will always need a wiping a rag or two. Following are checks that are common to the entire truck:

(1) Keep it Clean. Dirt, grease, oil, and debris only get in the way and may cover up a serious problem. Clean as you work and as needed. Use dry cleaning solvent (SD-2) (item 10, App D) on all metal surfaces. Use soap (item 21, App D) and water when you clean rubber or plastic material. Upholstery can be cleaned with soap and water and a clean, damp cloth (item 40, App D).

(2) Rust and Corrosion. Check truck body and frame for rust and corrosion. If any bare metal or corrosion exists, clean and apply a thin coat of oil (item 18, App D). Report it to your supervisor.

(3) Bolts, Nuts, and Screws. Check them all for obvious looseness, missing, bent, or broken condition. You can't try them all with a tool, but look for chipped paint, bare metal, or rust around bolt heads. If you find a bolt, nut, or screw you think is loose, tighten it or report it to your supervisor.

(4) Welds. Look for loose or chipped paint, rust, or gaps where parts are welded together. If you find a bad weld, report it to your supervisor.

(5) Electric Wires and Connectors. Look for cracked, frayed, or broken insulation, bare wires, and loose or broken connectors. Tighten loose connectors. Report any damaged wires to your supervisor.

(6) Hoses and Fluid Lines. Look for wear, damage, and leaks, and make sure clamps and fittings are tight. Wet spots show leaks, but a stain around a fitting or connector can also mean a leak. If a leak comes from a loose fitting or connector, tighten it. If something is broken or worn out, report it to your supervisor.

i. When you check for "operating condition," you look at the component to see if it's serviceable.

2.4.3 Cleaning Agents.

WARNING

- DO NOT use diesel fuel, gasoline, or benzene (benzol) for cleaning.
- DO NOT SMOKE when using cleaning solvent. NEVER USE IT NEAR AN OPEN FLAME. Be sure there is a fire extinguisher nearby and use cleaning solvent only in well-ventilated places. Flash point of solvent is 138°F (60°C).
- USE CAUTION when using cleaning solvents. Cleaning solvents evaporate quickly and can irritate exposed skin if solvents contact skin. In cold weather, contact of exposed skin with cleaning solvents can cause frostbite.

CAUTION

When cleaning underhood areas, engine must be COLD (same temperature as outside air). DO NOT point water or steam directly at any electrical connection. DO NOT point water stream directly at radiator fins. DO NOT use high pressure water supply system. Damage to engine, electrical system, and other components may result.

NOTE

Only use those authorized cleaning solvents or agents listed in Appendix D.

a. Cleaning Under Hood Areas.

(1) When using water to clean the engine compartment, always cover alternators and air cleaner inlet using waterproof material. For M1010 only, also cover air conditioner compressor. Use water pressure and volume similar to a standard household type water supply system (45-70 psi, 6.5-10.2 kPa).

FIGURE 2. Example of a PMCS introduction (sheet 2 of 3) (reference 5.5.2.1).

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(2) After cleaning, allow engine to air dry. Do not use compressed air to dry engine. Do not run engine to decrease drying time.

(3) Remove all component covers before starting engine.

b. Treating Mildewed Areas. Canvas that has mildewed can be cleaned by scrubbing with a dry brush. If it is necessary to use water to remove dirt, it should not be used until mildew has been removed. After removing mildew, examine fabric. Look for evidence of deterioration. If canvas has deteriorated, it should be replaced.

CAUTION

Keep cleaning solvents, gasoline, and lubricants away from rubber or soft plastic parts. They will deteriorate material.

c. Cleaning Rust or Grease. When cleaning grease buildup or rusty places, use a cleaning solvent (item 9, App D). Then apply a thin coat of light oil (item 18, App D) to affected area.

2.4.4 Leakage Definitions for Operator PMCS.

It is necessary for you to know how fluid leakage affects the status of the truck. Following are types/classes of leakage an operator needs to know to be able to determine the status of the truck. Learn these leakage definitions and remember - when in doubt, notify your supervisor.

CAUTION

- Equipment operation is allowable with minor leakages (Class I or II). Consideration must be given to fluid capacity in the item/system being checked/inspected. When in doubt, notify your supervisor.
- When operating with Class I or II leaks, continue to check fluid levels as required in your PMCS.
- Class III leaks should be reported immediately to your supervisor.

a. Class I - Seepage of fluid (as indicated by wetness or discoloration) not great enough to form drops.

b. Class II - Leakage of fluid great enough to form drops but not enough to cause drops to drip from item being checked/inspected.

c. Class III - Leakage of fluid great enough to form drops that fall from item being checked/inspected.

FIGURE 2. Example of a PMCS introduction (sheet 3 of 3) (reference 5.5.2.1).

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Table 2-1. Preventive Maintenance Checks and Services for (XXX). - Continued

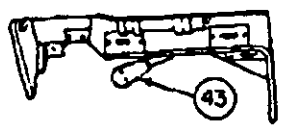
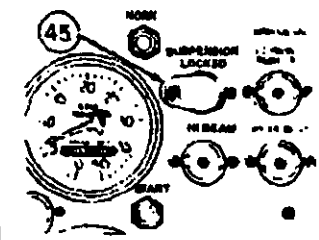
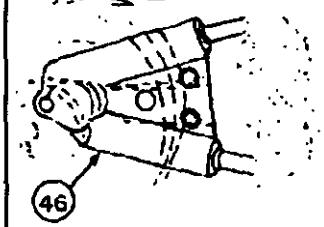
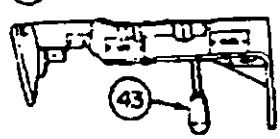
ITEM NO.	INTERVAL	LOCATION ITEM TO CHECK/ SERVICE	PROCEDURE	NOT FULLY MISSION CAPABLE IF:
23	Before/After	Suspension Lockout System	a. Put the lockout selector lever (43) in LOCK position. Make sure the SUSPENSION LOCKED warning light (45) comes ON and the lockout cylinders (46) lock against the road wheel arms. b. Put the selector lever (43) back in RELEASE position.  LOCK POSITION    RELEASE POSITION	SUSPENSION LOCKED warning light (45) does not come on or lockout cylinders (46) do not lock against road wheel arms.

FIGURE 3. Example of operator's PMCS table format (reference 5.5.2.3).

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Table 2-1. Preventive Maintenance Checks and Services for (XXX). - Continued

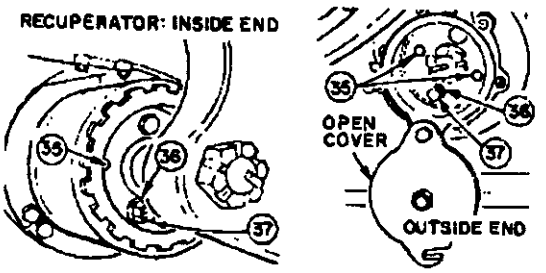
ITEM NO.	INTERVAL	LOCATION ITEM TO CHECK/ SERVICE	CREWMEMBER PROCEDURE	NOT FULLY MISSION CAPABLE IF:
7	Before	Recuperator	CREWMEMBER 1 **WARNING** Do not remove cap (37) when engine is hot. Check pins (35) before firing. Pins must not extend over 3/4 inch (1/8 inch is correct). If pins (35) do extend beyond 3/4 inch, remove cap (37) from valve (36) and add hydraulic fluid (item 5, App D) until pins extend 1/8 inch. **NOTE** The recuperator is charged with a pressure of 550 psi in the M109 and 700 psi in the M109A1. With temperature of 100° or over, 600 psi in the M109A1 is recommended. 	Pins (35) extend over 3/4 inch.

FIGURE 4. Example of operator's PMCS table format with crewmember grouping (reference 5.5.2.3c(2)).

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Table X-X. PMCS Mandatory Replacment Parts List
Semiannual (1500 Mile)

ITEM NO.	PART NUMBER	NSN	NOMENCLATURE	QTY
1	HC7500Y144	4330-01-217-8184	FILTER ELEMENT, FLUID (M977/M985 ONLY)	1
2	HD223	2940-01-132-4842	FILTER ELEMENT, FLUID	1
3	MS35338-46	5310-00-637-9541	WASHER, LOCK (ALL EXCEPT M984A1)	2
4	MS35802-3	2940-00-580-6283	FILTER ELEMENT, FLUID	1
5	S-268-1	5306-01-084-5390	BOLT, MACHINE (ALL EXCEPT M984A1)	2
6	WA110	5310-01-061-5302	WASHER, LOCK	4
7	10232C	5330-01-168-8707	GASKET (M978 ONLY)	1
8	11007	5330-01-046-1990	GASKET	1
9	1112310	5330-01-225-4803	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
10	1124510	5330-01-143-0135	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
11	1128242	5330-01-145-1112	PACKING, PREFORMED (M977/M985/M984A1 ONLY)	1
12	11350	5330-01-147-6003	PACKING, PREFORMED	1
13	1199478	5330-01-234-7625	GASKET	1
14	1199738	4330-01-192-7664	SCREEN, BY-PASS FILTER (M977/M985/M984A1 ONLY)	1
15	1300766	4330-01-232-8305	FILTER ELEMENT, FLUID (M984A1 ONLY)	1
16	1300767	4330-01-192-8832	FILTER ELEMENT, FLUID (M977/M985/M984A1 ONLY)	1
17	2020SM	4330-01-046-3399	FILTER ELEMENT, FLUID	1
18	2463HX	5310-01-054-5896	WASHER, FLAT (ALL EXCEPT M984A1)	2
19	25010778	2910-01-022-8183	FILTER ELEMENT, FLUID	1
20	4-1/2IN77/	5330-01-163-5849	GASKET (M978 ONLY)	2
21	45D020P6	5330-01-156-3764	PACKING, PREFORMED (M978 ONLY)	1
22	941107	4330-01-163-7326	FILTER ELEMENT, FLUID (M978/M983/M984/M985E1 ONLY)	1

FIGURE 5. Example of PMCS mandatory replacement parts list (reference 5.5.2.3.1).

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TM 9-2350-252-10PMC

CREW/OPERATOR DAILY PREVENTIVE MAINTENANCE CHECKLIST FOR HULL

FIGHTING VEHICLE, INFANTRY M2 AND M2A1 NSN XXXX-XX-XXX-XXXX (EIC XXX)

NOTICE

To effectively perform the tasks in this checklist, you must be experienced in using the PMCS table in TM 9-2350-252-10-1. The item numbers in this checklist are the same as those in the PMCS table.

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS

You can help improve this manual. If you find any mistakes, or if you know of a way to improve the procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms) direct to: (name and address of proponent). A reply will be furnished to you.

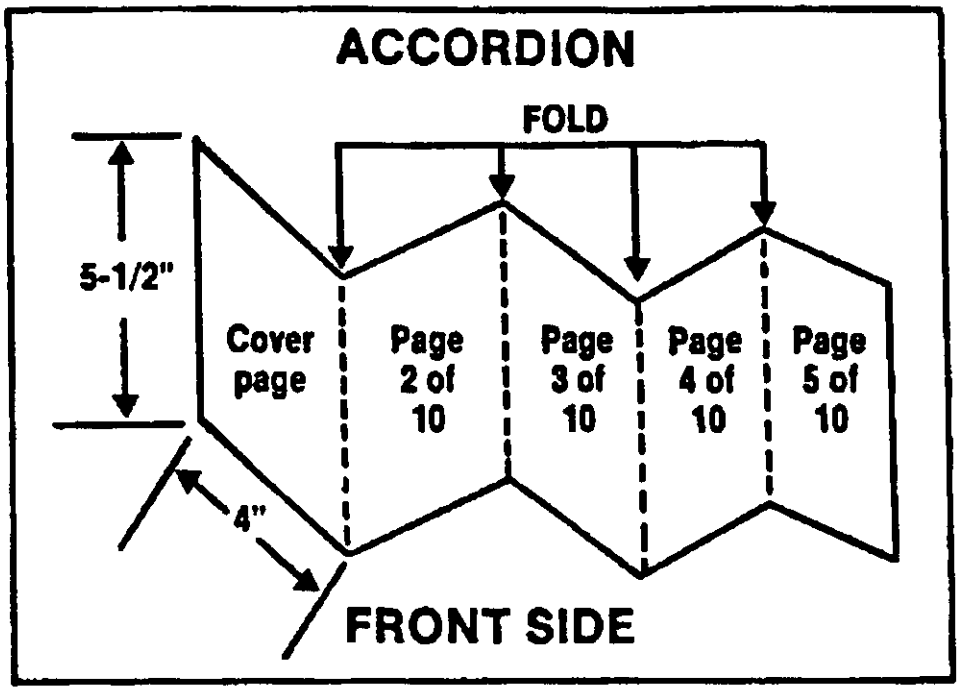
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HEADQUARTERS, DEPARTMENT OF THE ARMY

29 SEPTEMBER 1989

FIGURE 6. Example of a PMC cover (reference 5.5.2.4.2, 5.5.2.4.5).

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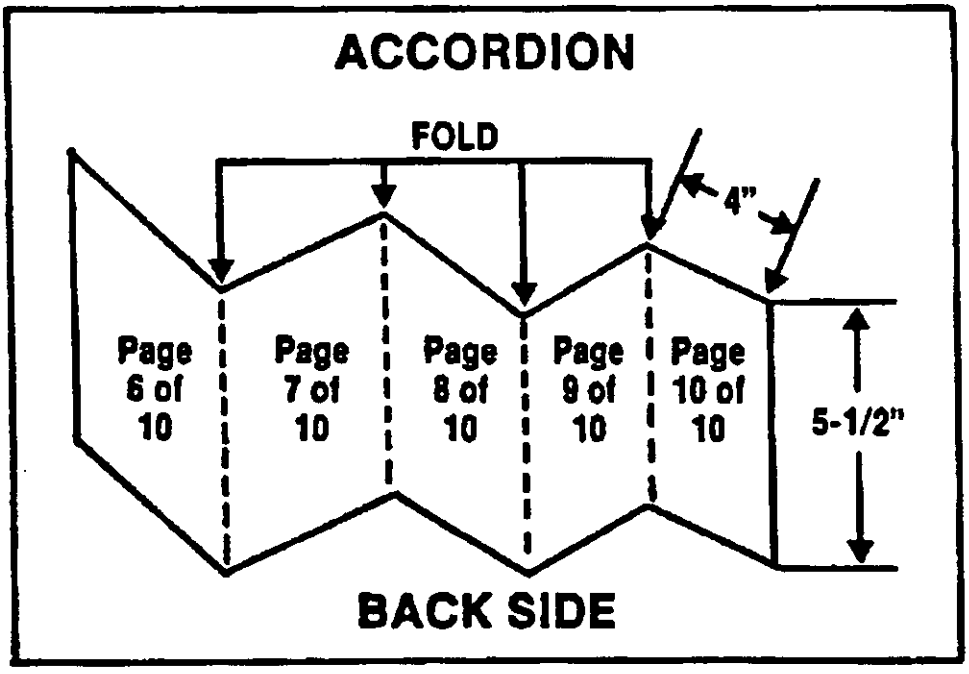


FIGURE 7. Example of an accordion foldout (reference 5.5.2.4.2c).

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BEFORE OPERATION CHECKS
DRIVER - CONTINUED

CAUTION

Vehicle must be on a level surface for proper oil or fluid checks.

*19. Check oil/fluid levels in these items:
  a. Engine
  b. Transfer gear case

WARNING

Carbon monoxide in exhaust can kill you. Make sure you have adequate ventilation and that the emergency brake is set before starting engine.

*20. Check radiator and heater hoses for leaks with the engine running.

*21. Check fuel lines for leaks.

22. Turn engine off and then drain fuel filter contamination.

GUNNER

23. Check the deck and make sure all items are secured.

*24. Check the machine gun mount for proper operation.

25. Check the commander's cupola.

NOTE

An asterisk indicates a Not Fully Mission Capable (NFMC) fault that must be reported on DA Form 2404 or other maintenance form prescribed by your command.

FIGURE 8. Example of a PMC inside page identifying crewmembers (reference 5.5.2.4.3d).

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TM 9-2350-255-PCL

COMBAT VEHICLE
PRECOMBAT CHECKLIST FOR:
TANK, COMBAT, FULL TRACKED, 105MM
GUN, M1, IPM1, GENERAL ABRAMS
NSN 2350-01-061-2445 (EIC XXX)
NSN 2350-01-138-8738 (EIC YYY)

WARNING

THIS CHECKLIST IS TO BE USED ONLY WHEN AUTHORIZED BY THE COMMANDER. IT IS NOT INTENDED TO BE USED IN PLACE OF THE PMCS.

NOTE

THIS CHECKLIST CONTAINS CHECKS TO BE PERFORMED BEFORE COMBAT TO ASSURE VEHICLE IS MISSION CAPABLE. THE CHECKLIST SEQUENCE NUMBERS HEREIN CORRESPOND TO THOSE PMCS CHECKS IN TM 9-2350-255-10-1. REPORT CHECK RESULTS TO YOUR IMMEDIATE LEADER.

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By Order of the Secretary of the Army:

CARL E. VUONO
General, United States Army
Chief of Staff

Official:

THOMAS F. SIKORA
Brigadier General, United States Army
The Adjutant General

Special Distribution:

To be distributed in accordance with DA Form 12-37, Block 1088, four per account, Operator Maintenance Requirements for Tank, Combat, Full-Tracked, 105MM, M1 Abrams.

DEPARTMENT OF THE ARMY

8 MARCH 1990

FIGURE 9. Example of PCL cover page (reference 5.5.2.5.2b, 5.5.2.5.3.1).

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1,4.	**VEHICLE EXTERIOR:** Walk Around And Check For Fluid Leaks, Obvious Damage, Tampering, And Missing Parts That Prevent Operation.
5.	**ENGINE COMPARTMENT:** Check Inside Engine Compartment For Leaks And Damage.
20.	**ENGINE OIL CHECK:** Check Oil Level.
14.	**COOLANT LEVEL:** Check Coolant Level.
2, 16-17.	**FIRE EXTINGUISHERS:** Check Seals And Bottle Pressures.
40-46.	**ENGINE INSTRUMENTS:** Start Engine; Check Instruments.
37.	**COMMUNICATION SYSTEM:** Radio And Intercom Operational.
34, 56, 60.	**GUNNER CONTROL HANDLES:** Gun Elevation And Turret Traverse Operational.
	FIRE CONTROL SYSTEM:
28.	A. Nightsight Check.
12, 59.	B. M60 Gun Check.
	C. TOW System Self-Test.

2

FIGURE 10. Example of a checklist for a TM with one PMCS table (reference 5.5.2.5.3.2a).

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TABLE 2-1

1. **VEHICLE EXTERIOR:** Walk Around And Check For Fluid Leaks, Obvious Damage, Tampering, And Missing Parts That Prevent Operation.

21,40. **FIRE EXTINGUISHER SYSTEM:** Check Bottle Pressure.

43. **ENGINE/INSTRUMENTS:** Start Engine; Check Instruments.

******NOTE: TURN TIS TO STANDBY MODE******

24. **COMMUNICATION SYSTEM:** Radio and Intercom Operational.

TABLE 2-2

1,2,12. **MACHINE GUNS:** Mounted And Operational.

6. **TURRET AMMUNITION STOWAGE:** Compartment Doors Operational.

9. **COMMANDER/GUNNER POWER CONTROL HANDLES:**
A. Gun Elevation/Turret Traverse.
B. Aux Hyd Gauge Reads 1150-1700.
C. Electrical Triggers And Blasting Device Operational.

13. **FIRE CONTROL SYSTEM:**
A. TIS Checkout.
B. Computer Self-Test.
C. Computer Data Check.
D. GPS Function Check.

4.1. **STUB BASE DEFLECTOR INSTALLED.**

2

FIGURE 11. Example of a checklist for a TM with more than one PMCS table (reference 5.5.2.5.3.2a).

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TM 9-2350-284-10-1

2,3. **VEHICLE EXTERIOR:** Walk Around And Check For Fluid Leaks, Obvious Damage, Tampering, And Missing Parts That Prevent Operation.

******NOTE: LOWER RAMP******

4. **HULL DRAIN PLUG FRONT:** Close Front Drain Plug.

12. A,B. **FIRE EXTINGUISHER SYSTEM FRONT:** Check Bottle Pressure.

15-17. **FIRE EXTINGUISHER SYSTEM REAR:** Check Bottle Pressure.

4,15. **HULL DRAIN PLUG REAR:** Close Rear Drain Plug.

7-8. **ENGINE/INSTRUMENTS:** Start Engine; Check Instruments.

TM 9-2350-284-10-2

******POWER UP ISU******

13. **COMMUNICATION SYSTEM:** Radio and A,B. Intercom Operational.

23. **NBC SYSTEM CHECK.**

17.A-B, **COMMANDER/GUNNER POWER CONTROL**
26. **HANDLES:** Gun Elevation And Turret Traverse Operational.

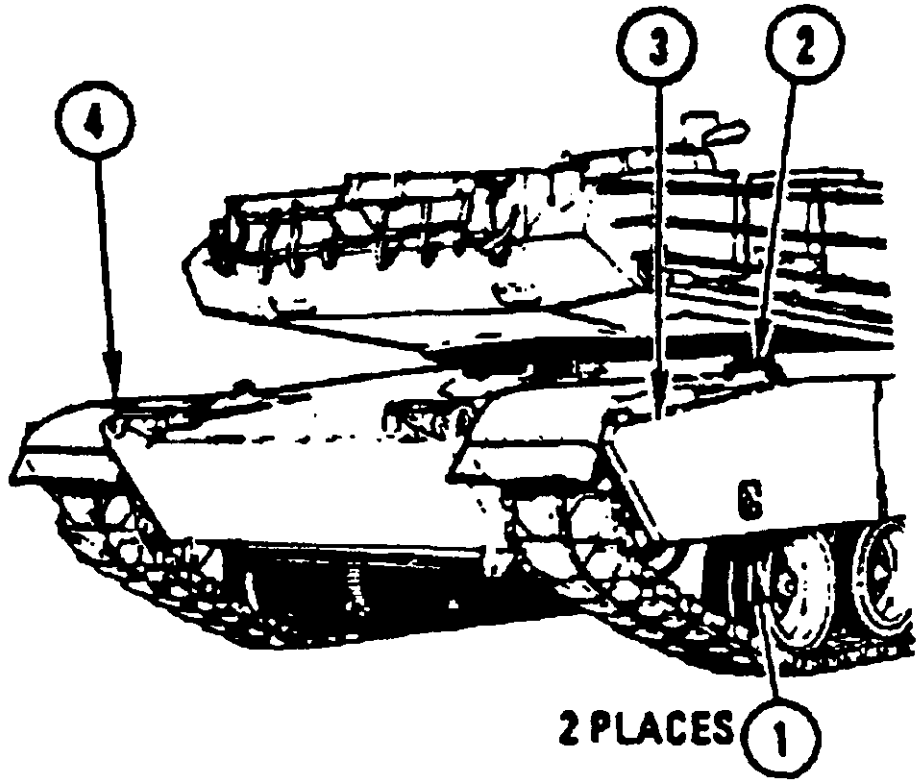
FIRE CONTROL SYSTEM:
29,30. A. ISU Checkout.
33. B. 25MM Gun Checkout.
35. C. Coaxial Machine Gun Function Check.
D. Tow System Self-Test.

2

FIGURE 12. Example of a checklist for a TM with more than one volume (reference 5.5.2.5.3.2a).

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2.3 ASSEMBLY AND PREPARATION FOR USE.



a. Driver's Station. Prepare station for operation (Enter driver's station.) as follows:

WARNING

If main gun is to the front of tank, climb on tank from left side only to keep from getting in laser or machine gun firing path.

NOTE

Main gun must be over rear deck to enter driver's station from turret.

(1) Climb on tank using track skirt step (1) and hand hold (2) at left front (3), or using skirt step (1) at right front (4) of hull.

(2) Unlock and open loader's hatch. (See paragraph 2.5.)

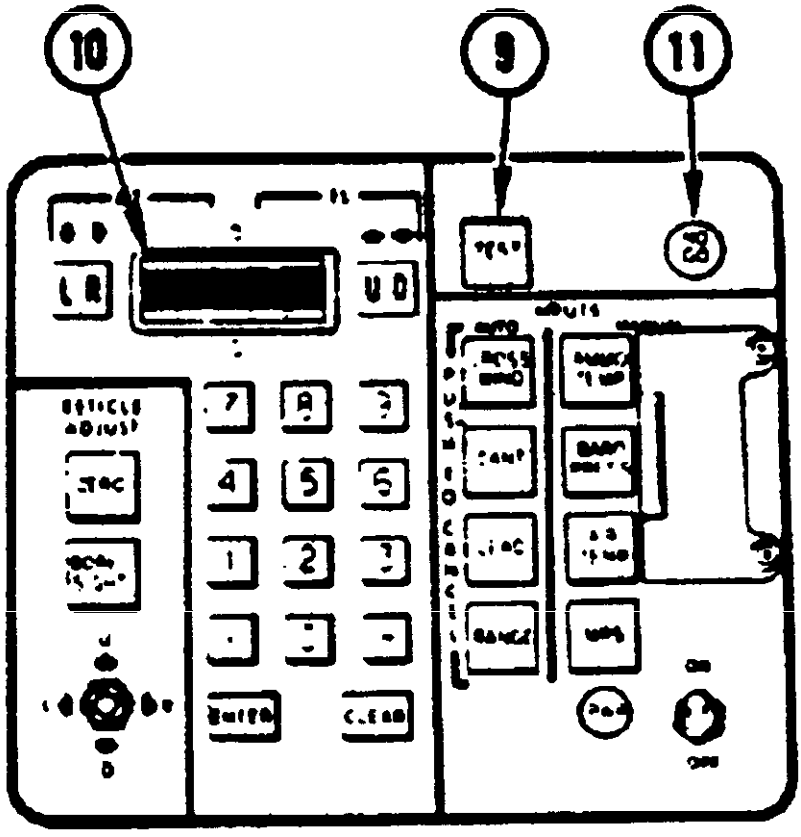
(3) Enter tank through loader's hatch feet first. Step on loader's seat post, then on turret floor.

(4) Make sure turret traverse lock is set to LOCKED. (See paragraph 2.5.)

FIGURE 13. Example of operation procedure format (sheet 1 of 2) (reference 5.5.3).

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2.4 OPERATING PROCEDURE FOR GUNNER'S STATION BALLISTIC COMPUTER SELF-TEST - Continued.



j. Push and release TEST button (9). Test button (9) lights and stays lit during entire test.

k. Watch display (10) and NO GO light (11). When display (10) and/or NO GO light (11) comes on, do step below for that display.

NOTE

During the self test, numbers will appear in display (10) as part of the test. These numbers will remain in display (10) for 4 seconds, and then go out. Numbers should be between 9.5 and 10.5. If not, notify unit maintenance.

All failure numbers 1 through 8 are listed in the general computer operating instructions on inside of CCP door.

The self test failure numbers and words 'PASS' or 'FAIL' will go off automatically ten seconds after appearing in display (10).

(1) If NO GO light (11) comes on and a number 2, 3, 4, or 8 appears in display (10), continue to squeeze palm switches and go to step l.

FIGURE 13. Example of operation procedure format (sheet 2 of 2) (reference 5.5.3).

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NOTE
DETAILED DECON PROCEDURES CAN BE FOUND IN:
FM 3-3, FM 3-4, AND FM 3-5.

GENERAL:

THE FOLLOWING EMERGENCY PROCEDURES CAN BE PERFORMED UNTIL FIELD NBC DECON FACILITIES ARE AVAILABLE. TANK COMMANDER WILL SUPERVISE, ASSIGN CREW DUTIES, AND ASSIST THE SUPPORTING NBC UNIT. FOR LOCATION OF DECON EQUIPMENT MOUNTED ON VEHICLE, SEE PAGE XX-XX.

EMERGENCY PROCEDURES:

IF NBC ATTACK IS KNOWN OR SUSPECTED, MASK AT ONCE AND CONTINUE MISSION. IF INSIDE, DO NOT LEAVE TANK. IF OUTSIDE, FOLLOW DECON PROCEDURES BELOW TO AVOID TAKING CONTAMINATION INTO THE TANK. DO NOT UNMASK UNTIL TOLD TO DO SO.

NUCLEAR DECONTAMINATION -

BRUSH FALLOUT FROM SKIN, CLOTHING, AND EQUIPMENT WITH AVAILABLE BRUSHES, RAGS, AND TREE BRANCHES. WASH SKIN AND HAVE RADIATION CHECK MADE AS SOON AS TACTICAL SITUATION PERMITS. (YOU CAN FIND INSTRUCTIONS FOR THE CHECK IN FM XX-XXX.)

BIOLOGICAL DECONTAMINATION -

THE TANK CREW HAS NO METHOD TO DETECT OR DECON BIOLOGICAL AGENTS. REMAIN MASKED AND CONTINUE MISSION UNTIL TOLD TO UNMASK.

FIGURE 14. Example of NBC presentation (sheet 1 of 2) (reference 5.5.4.4).

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EMERGENCY PROCEDURES: - CONTINUED

CHEMICAL DETECTION AND DECONTAMINATION -

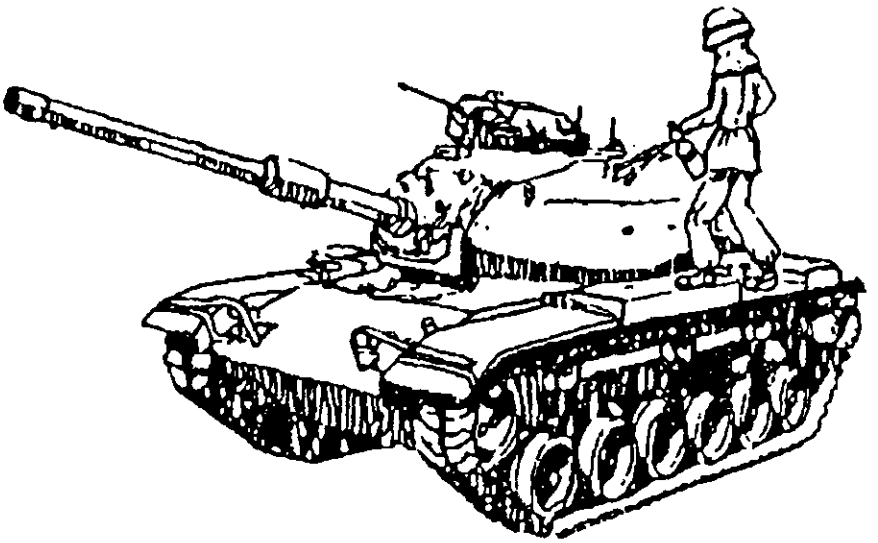
WARNING
DO NOT USE DECONTAMINATION SPRAY ON PERSONNEL. IT COULD CAUSE PERSONAL INJURY.

USE M8 PAPER FROM THE M256 CHEMICAL AGENT DETECTOR KIT OR M9 PAPER TO DETERMINE IF LIQUID AGENT IS PRESENT ON THE VEHICLE SURFACE.

IF EXPOSURE TO LIQUID AGENT IS KNOWN OR SUSPECTED, CLEAN EXPOSED SKIN, CLOTHING, AND PERSONAL GEAR, IN THAT ORDER, USING M258A1 KIT. USE THE BUDDY SYSTEM. WASH EXPOSED SKIN AND THOROUGHLY DECONTAMINATE AS SOON AS TACTICAL SITUATION PERMITS.

IF THE M8 OR M9 PAPER INDICATES THAT LIQUID CHEMICAL AGENT IS PRESENT ON THE VEHICLE SURFACE, USE THE ABC-M11 DECON APPARATUS FOR PARTIAL DECON OF VEHICLE. USE LOADER'S HATCH FOR EXIT/ENTRY. AVOID GETTING LIQUID AGENT INTO CREW COMPARTMENT. SPRAY ONLY SURFACES THAT WILL BE TOUCHED GETTING IN AND OUT OF TANK.

DECON PROCEDURES TAKE TIME. DO AS MUCH AS YOU CAN BASED ON THE TACTICAL SITUATION.



LOADER USING M11 DECON APPARATUS TO SPRAY AN ENTRY/EXIT PATH

FIGURE 14. Example of NBC presentation (sheet 2 of 2) (reference 5.5.4.4).

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Table 3-1. Troubleshooting - Continued.

MALFUNCTION TEST OR INSPECTION CORRECTIVE ACTION
ARMAMENT - 155-MM HOWITZER CANNON - Continued
Step 2. Check to see if extractor is bent or broken. Repair/replace (paragraph 3.33).
23. EXCESSIVE BLOWBACK AROUND BREECHBLOCK.
Step 1. Check to see if split rings were installed properly. Rotate so split rings are 180° apart (paragraph 3.50).
Step 2. Check to see if obturator pad/split rings are worn. Remove obturator (paragraph 3.49) and notify unit maintenance.
24. BREECH MECHANISM DOES NOT OPERATE FREELY.
Step 1. Check to see if breech threads or components are dirty or burred on breechblock/ring. Clean (paragraph 3.10).
Step 2. Inspect for insufficient lubrication. Lubricate (Appendix G).
Step 3. Check to see if breech threads or components are damaged or worn. Notify unit maintenance.
25. GASES ACCUMULATE IN CAB AFTER FIRING (M109).
Step 1. Check to see if evacuator valve assemblies are dirty. Clean (paragraph 3.12).
Step 2. Check to see if evacuator drain plug is loose or missing. Tighten/replace (paragraph 3.42).
Step 3. Check to see if the evacuator valve caps are loose. Tighten (paragraph 3.42).
Step 4. Check to see if rubber packing rings in chamber evacuator are worn. Notify unit maintenance.

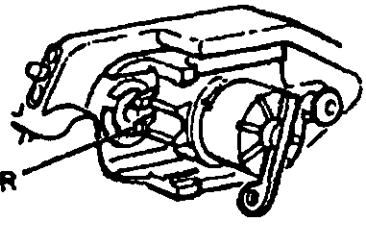


FIGURE 15. Example of a typical troubleshooting table (reference 5.6.2.1).

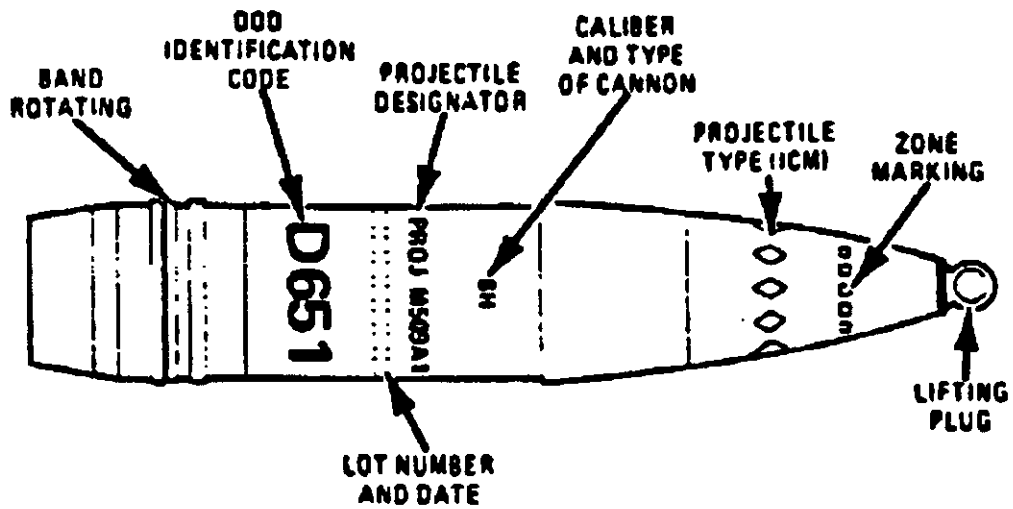
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MALFUNCTION INDEX

	Troubleshooting Procedure (Paragraph)
COOLING SYSTEM	
Radiator	
Boils over	3.12
Leaks	3.17
Temperature Gage	
No indication	3.29
Runs cold	3.29
Runs hot	3.29
ENGINE	
Misses	3.34
Overheats	3.72
Won't start	3.72
EXHAUST SYSTEM	
Excessive smoke	3.80
Water vapor	3.80

FIGURE 16. Example of a troubleshooting malfunction index (reference 5.6.2.3).

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MARKING OF 8-INCH DPICM, M509A1 PROJECTILE

FIGURE 17. Example of an ammunition identification illustration (reference 5.8).

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Table X-X. Authorized Projectile Fuze Combinations for 8-Inch Howitzer, SP, M110A2 Cannon M201A1.

TYPE AND MODEL NUMBER OF PROJECTILE	FUZE										
	PD	MT			MTSQ			PROX(VT)		ET	
	M739 SERIES	M557	M572	M565	M564	M557 SERIES	M582 SERIES	M728	M732	M762	M767
Agent GB, VX, M426	X	X	X					X 4	X 4		
HE, M106 (Shallow Cavity)	X	X	X		X 3		X		X		X
HE, M106 (Deep Cavity)	X	X	X		X 3		X	X 2	X		X
HE, M404 ICM				X		X				X	
HE, M509A1 ICM						X				X	
HERA, M650 (Rocket-On)	X	X	X								X
HERA M650[1] (Rocket-Off)	X	X	X		X 3		X		X		X

WARNING: 1-DO NOT FIRE THE M650 PROJECTILE IF THE OBTURATING BAND IS MISSING OR BROKEN. IF THE BAND IS DISPLACED AND CAN BE REPOSITIONED AND REMAIN IN THE GROOVE, THE PROJECTILE CAN BE FIRED.

NOTE: 2-AUTHORIZED, REQUIRES REMOVAL OF SUPPLEMENTARY CHARGE.

3-FUZE, MTSQ, M564 IS RESTRICTED FROM FIRING WITH ZONE 9 M188A1 PROPELLING CHARGE.

4-M728 AND M732 FIRED ONLY WITH "VX" PROJECTILE AND ONLY IN COMBAT EMERGENCY.

FIGURE 18. Example of a typical ammunition table (sheet 1 of 3) (reference 5.8).

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Table X-X. Authorized Propelling Charges Combinations for 8-Inch Howitzer, SP, M110A2 with Cannon M201A1.

TYPE AND MODEL NUMBER OF PROJECTILE	M1 ZONE					M2 ZONE			M188 ZONE	M188A1 ZONE	
	1	2	3	4	5	5	6	7	8	8	9
Agent GB, VX, M426	X	X	X	X	X	X	X	X	X	X	X
HE, M106 (Shallow Cavity)	X	X	X	X	X	X	X	X	X	X	X 2
HE, M106 (Deep Cavity)	X	X	X	X	X	X	X	X	X	X	X 2
HE, M404, ICM	X	X	X	X	X	X	X	X			
HE, M509A1, ICM	X	X	X	X	X	X	X	X	X	X	X
HERA, M650 (Rocket-On)								X	X	X	X
HERA M650[1] (Rocket-Off)	X	X	X	X	X	X	X	X	X	X	X 2

WARNING: 1-DO NOT FIRE THE M650 PROJECTILE IF THE OBTURATING BAND IS MISSING OR BROKEN. IF THE BAND IS DISPLACED AND CAN BE REPOSITIONED AND REMAIN IN THE GROOVE, THE PROJECTILE CAN BE FIRED.

NOTE: 2-FUZE, MTSQ, M564 IS RESTRICTED FROM FIRING WITH ZONE 9 M188A1 PROPELLING CHARGE.

FIGURE 18. Example of a typical ammunition table (sheet 2 of 3) (reference 5.8).

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Table X-X. Model Number and Color Coding of Projectiles for 8-inch Howitzer, M110A2.

TYPE AND MODEL NO. OF PROJECTILE	NEW MANUFACTURE			OLD MANUFACTURE		
	COLOR OF PROJECTILE	NO./COLOR OF BANDS	MARKING	COLOR OF PROJECTILE	NO./COLOR OF BANDS	MARKING
**Agent (GB) w/burster, M426	Gray	3/Green 1/Yellow	Green	Gray	1/Green	Green
**Agent (VX) w/burster, M426	Gray	3/Green 1/Yellow	Green	Gray	2/Green	Green
HE, (ICM) M404	Olive Drab	*Diamonds	Yellow	Olive Drab	None	Yellow

* Row of yellow diamonds between nose and bourrelet of projectile.

** Renovated or newly manufactured projectile (post 1976) will be marked with one green band and, if burster, one yellow band.

FIGURE 18. Example of a typical ammunition table (sheet 3 of 3) (reference 5.8).

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APPENDIX A

REFERENCES

A-1 SCOPE.

This appendix lists all forms, field manuals, technical manuals, and miscellaneous publications referenced in this manual.

A-2 FORMS.

Accident Identification Card ..... DD Form 518

Equipment Inspection and Maintenance Work Sheet ..... DA Form 2404

Maintenance Request ..... DA Form 2407

Product Quality Deficiency Report ..... SF 368

Uncorrected Fault Record ..... DA Form 2408-14

Vehicle Accident Report ..... SF 91

A-3 FIELD MANUALS.

Basic Cold Weather Manual ..... FM 31-70

Northern Operations ..... FM 31-71

Operation and Maintenance of Ordnance Materiel in
Cold Weather (0° to -65° F). ..... FM 9-207

A-4 TECHNICAL MANUALS.

Lance Missile System, Destruction of Equipment ..... TM 750-244-4-1

Operator's Manual: LANCE Guided Missile System ..... TM 9-1425-485-10-2

Operator's, Organizational, DS, and GS Maintenance Manual:
Storage Batteries, Lead-Acid Type ..... TM 9-6140-200-14

Organizational Maintenance Manual: Carrier
Guided Missile Equipment, Self-Propelled: M667 ..... TM 9-1450-485-20-2

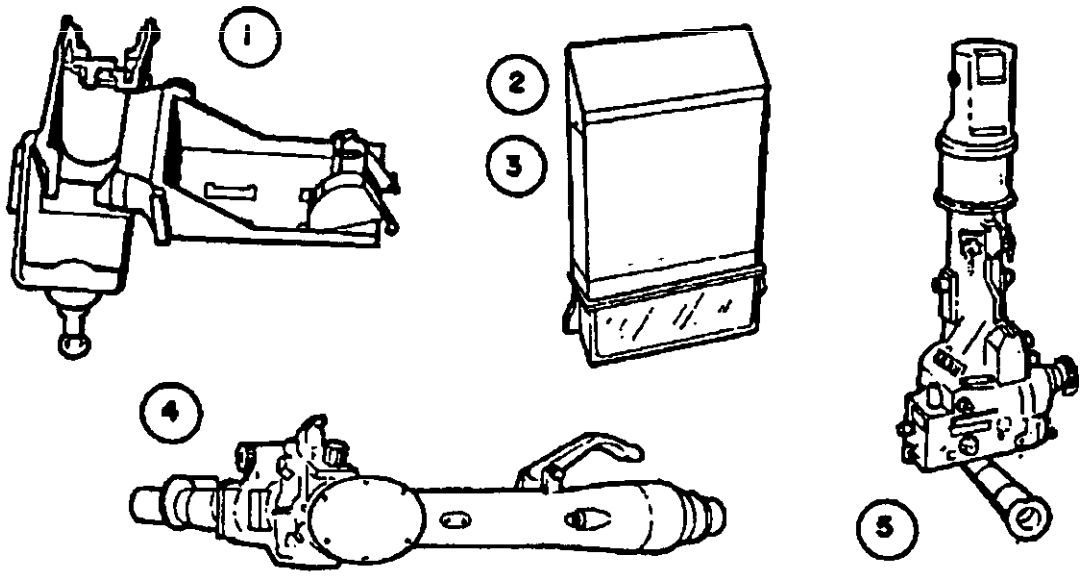
A-5 MISCELLANEOUS PUBLICATIONS.

Lubrication Order: Carrier, Guided Missile
Equipment, Self-Propelled: M667 ..... LO 9-1450-485-12

FIGURE 19. Example of references appendix (reference 5.10.1).

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SECTION II. COMPONENTS OF END ITEM

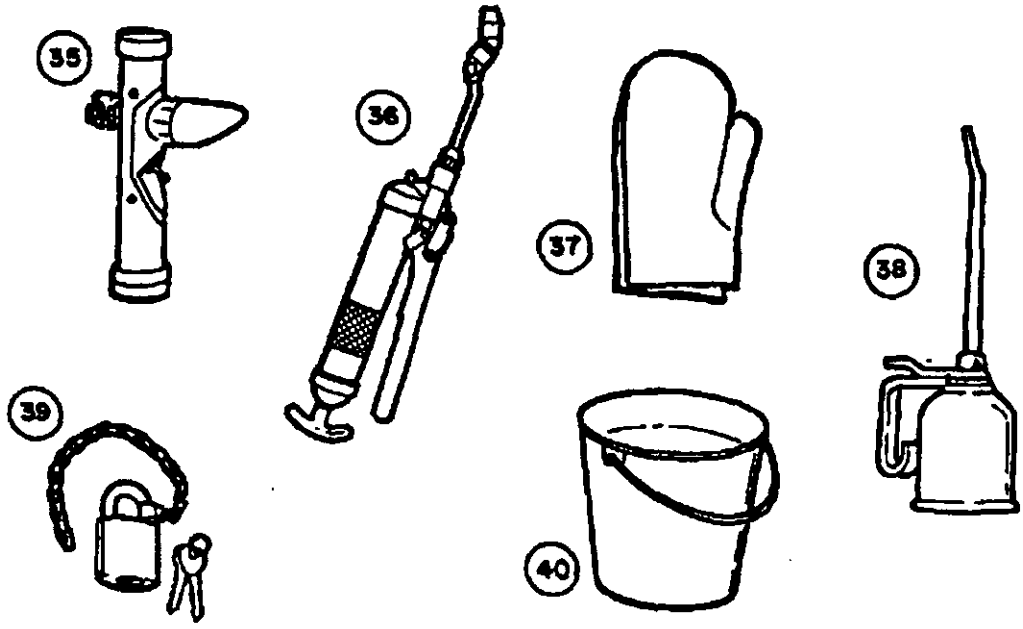


(1) ILLUS NUMBER	(2) NATIONAL STOCK NUMBER	(3) DESCRIPTION CAGEC AND PART NUMBER	USED ON CODE	(4) U/M	(5) QTY RQR
1	1005-00-706-8880	MOUNT, MACHINE GUN: cal. .50 (in mount on cupola) (19204) 7068880	PAA	EA	1
2	1240-00-344-4643	PERISCOPE: M27 (chief of section) (stowage box cab wall) (19200) 7633132	PAA	EA	1
3	1240-00-509-2743	PERISCOPE: M45 (driver's) (stowage box driver's compartment) (19200) 8213430	PAA	EA	3
4	1240-00-491-9676	TELESCOPE, ELBOW: M118CA1 (in mount M146) (19200) 10559855	PAB	EA	1
		OR			
	1240-00-819-4520	M118C (19200) 10512985	PAC	EA	1
5	1240-00-864-2930	TELESCOPE, PANORAMIC M117 (in mount M145 or telescope box) (19200) 7660400	PAA	EA	1

FIGURE 20. Example of a components of end item list (reference 5.10.2.2).

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SECTION III. BASIC ISSUE ITEMS



(1) ILLUS NUMBER	(2) NATIONAL STOCK NUMBER	(3) DESCRIPTION CAGEC AND PART NUMBER	USED ON CODE	(4) U/M	(5) QTY RQR
35	1290-00-535-7629	LIGHT, AIMING POST: M14 (19200) 7197188	PAA	EA	2
36	4930-00-766-3545	LUBRICATING GUN, HAND: high pressure (in tool bag) (36251) 102758	PAA	EA	1
37	8415-00-266-8843	MITTENS, CLOTH: (pair) M1942 (in oddment tray) (19207) 11655982	PAA	PR	2
38	4930-00-262-8868	OILER, HAND: steel, pump type, 1 pt, spout 9 lg (in left cab door stowage box) (72798) 328	PAA	EA	1
39	5340-00-682-1508	PADLOCK: set 1-3/4 w clevis-includes 1 lock & 2 keys (on hull rear door) (80063) SMD555531-1	PAA	EA	1
40	7240-00-160-0455	PAIL, UTILITY: 14-qt capacity (on vehicle floor) (81348) RRP35	PAA	EA	1

FIGURE 21. Example of a basic issue items list (reference 5.10.2.3, 5.10.2.4).

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SECTION II. ADDITIONAL AUTHORIZED ITEMS

(1) NATIONAL STOCK NUMBER	(2) DESCRIPTION CAGEC & PART NUMBER	USED ON CODE	(3) U/M	(4) QTY RECM
5935-00-322-8959	Adapter, Connector (19027) 11677570	ALL	EA	2
4730-00-808-5089	Adapter, Straight Pipe (81361) B5-19-1676-2	210	EA	23
4730-00-808-5090	Adapter, Straight Pipe (81361) B5-19-1676-1	210	EA	2
6665-00-935-6955	Alarm, Chemical Agent (81361) C5-15-8803	ALL	EA	1
5110-00-293-2336	Ax: Single Bit, 4 lb. Head Weight, 36-1/2" Long, Type 1, Class 1, Design A (19207) 6150925	209	EA	1
2540-00-670-2459	Bag Assembly, Pamphlet Cotton Duck, 3" X 9-1/4" X 11-1/4" (19207) 11676920	ALL	EA	1
5140-00-473-6256	Bag, Tool, Cotton Duck, 10" X 20" with Flap (34623) 11655979	ALL	EA	1
2540-00-791-3343	Block, Chock (19207) 10860840	ALL	EA	2
5340-00-595-5208	Bracket, Angle (81361) B5-19-1831	210	EA	7
2590-00-148-7961	Cable, NATO Slave, 20' with Adapter, Connector, 2 Prong (19207) 11682379-1	ALL	EA	1
6150-01-022-6004	Cable, NATO Slave with End Connectors (19207) 11682336-1	ALL	EA	1
2590-00-398-6527	Cable, 20' without End Connectors (19207) 11682337-1	ALL	EA	1
2540-01-185-8306	Chain, Tire, 15" (16457) 3210	209	SE	1
2540-00-528-7360	Chain, Tire, 16" (72671) V2812	ALL EXC 209	SE	1
4730-00-554-7208	Clamp, Hose (98441) 10181-20	210	EA	16
4730-00-554-7208	Clamp, Hose (96906) MS 22064-5	210	EA	30
1025-00-007-9453	Connector, Orifice (81361) B5-19-1829	210	EA	7
2510-00-567-0128	Connector, Slave Cable End (19207) 11682338	ALL	EA	2

FIGURE 22. Example of format for additional authorized items (reference 5.10.3.2).

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SECTION II. EXPENDABLE AND DURABLE ITEMS LIST

(1) ITEM NUMBER	(2) LEVEL	(3) NATIONAL STOCK NUMBER	(4) ITEM NAME, DESCRIPTION CAGEC, PART NUMBER	(5) U/M
1	C	6850-00-174-1806	Antifreeze: Arctic 55-Gallon Drum (81349) MIL-A-11755	gal.
2	C	6850-00-181-7929	Antifreeze: Ethylene Glycol, Inhibited, Heavy Duty, Single Package, 1-Gallon Can (81349) MIL-A-46153	gal.
3	C	6850-00-181-7933	Antifreeze: Ethylene Glycol, Inhibited, Heavy Duty, Single Package, 5-Gallon Can (81349) MIL-A-46153	gal.
4	C	6850-00-181-7940	Antifreeze: Ethylene Glycol, Inhibited, Heavy Duty, Single Package, 55-Gallon Drum (81349) MIL-A-46153	gal.
5	C	9150-01-102-9455	Brake Fluid: Silicone, Automotive, All Weather Operational and Preservative, 1-Gallon Can (81349) MIL-B-46176	gal.
6	C	9150-01-123-3152	Brake Fluid: Silicone, Automotive, All Weather Operational and Preservative, 5-Gallon Can (81349) MIL-B-46176	gal.

FIGURE 23. Example of an expendable and durable items list format (reference 5.10.4).

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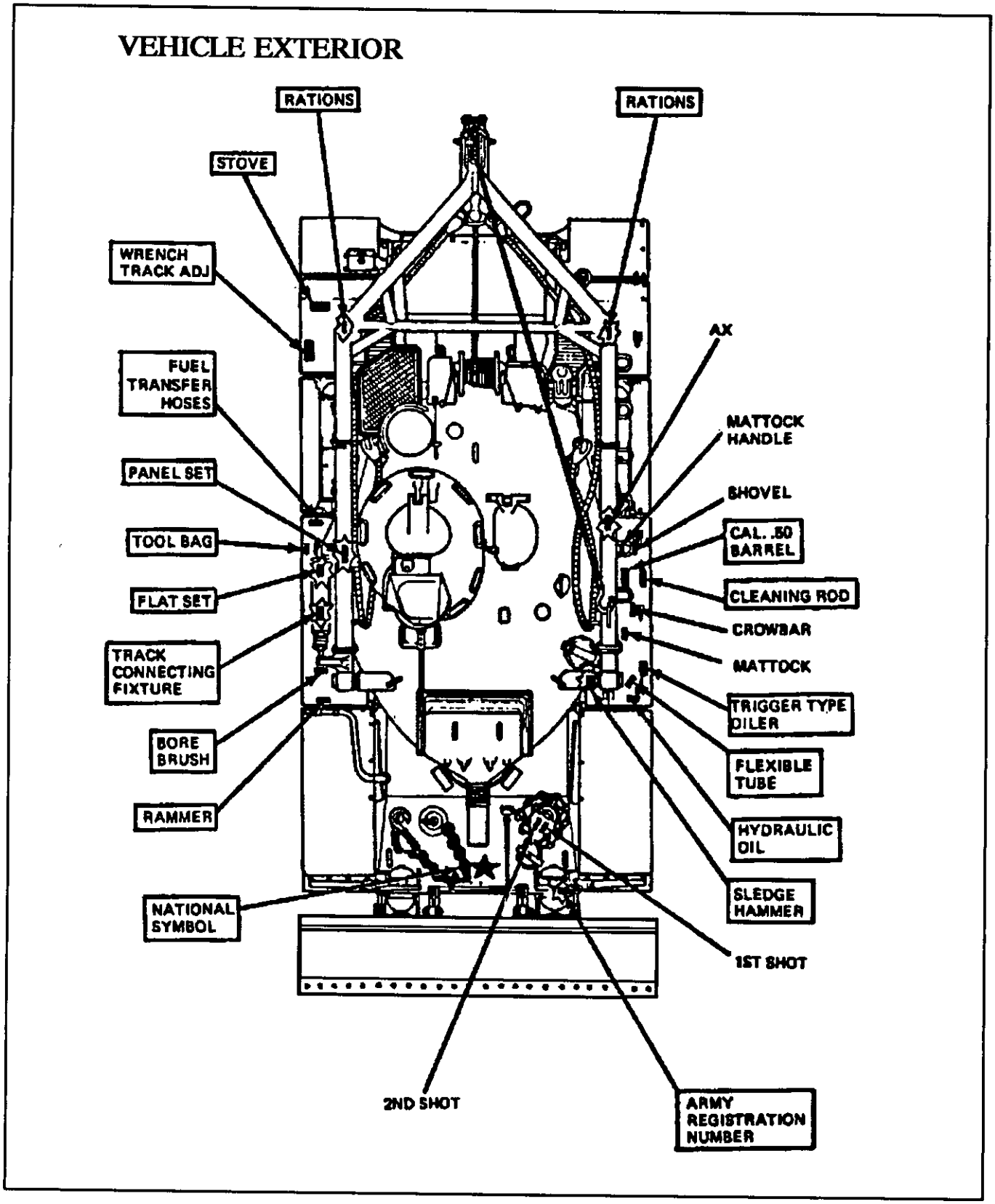
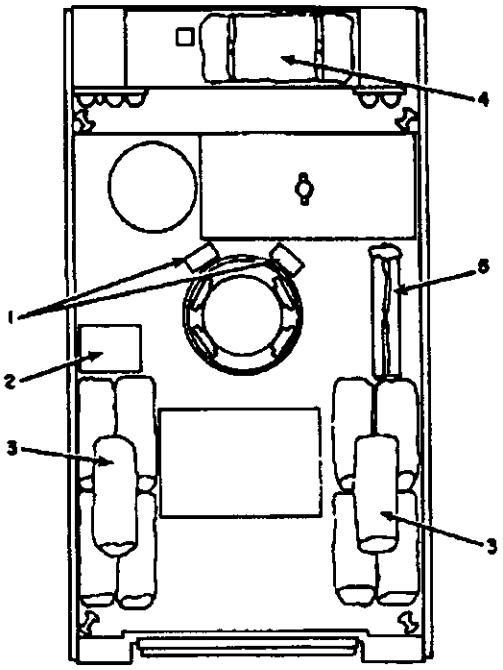


FIGURE 24. Example of stowage and decal/data plate guide appendix (reference 5.10.5).

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EXTERNAL TOP VIEW (TACTICAL)



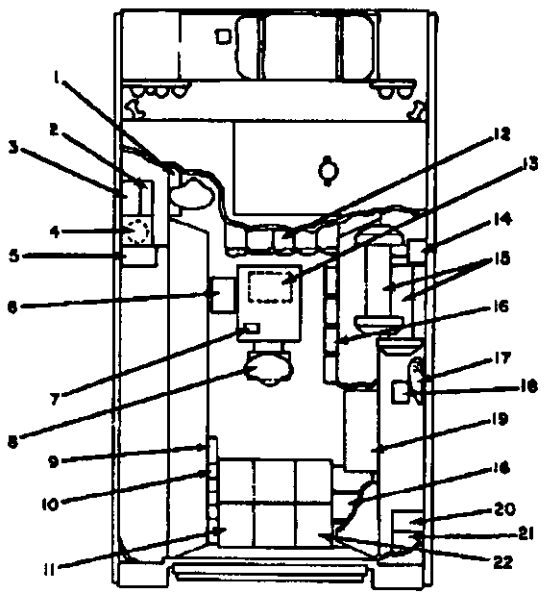
M113A1 CARRIER

LOAD PLAN	
NO.	ITEM
1	AMMO BOXES, 50 CAL (2)
2	C-RATION CASE (2)
3	DUFFEL BAGS W/BEDROLLS (10)
4	CAMOUFLAGE SCREEN
5	CAMOUFLAGE SUPPORT SYSTEM

FIGURE 25. Example of a tactical on-vehicle loading plan (reference 5.10.6).

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INTERNAL TOP VIEW (NON-TACTICAL)



M113A1 CARRIER

LOAD PLAN	
NO.	ITEM
1	MAP CANISTER
2	TELEPHONE SET, TA-1/PT
3	TELEPHONE SET, TA-312/PT
4	CABLE TELEPHONE, WD-1/TT, DR-8, 1320 FT
5	BINOCULAR, MOD CONST MIL SCALE RETICLE 7K50 MM W/F

FIGURE 26. Example of a non-tactical on-vehicle loading plan (reference 5.10.6).

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INDEX

	PARAGRAPH	PAGE
Abbreviations	4.2.5	10
Acquisition requirements	6.4	32
Acronyms used in this standard	3.1	4
Additional authorization list (AAL) appendix	5.10.3	28
Additional authorization list (AAL) items	3.2	5
Advertising	4.2.10	10
After operation checks	3.3	5
Alphabetical index	5.10.10	31
Appendixes	5.10	26
APPLICABLE DOCUMENTS	2.	2
Applicability	1.3	1
Arrangement of operator's TM (-10)	5.3	11
Arrangement of procedures	5.6.2.1	25
Assembly and preparation for use	5.5.3.2	21
Auxiliary equipment	3.4	5
Basic issue items (BII)	3.5	5
Battlefield damage assessment and repair (BDAR)	3.6	5
Before operation checks	3.7	5
Caution	3.8	5
Chapter 1 - introduction	5.4	12
Chapter 2 - operating instructions	5.5	12
Chapter 3 - operator maintenance instructions	5.6	24
Chapter 4 - auxiliary equipment maintenance	5.7	26
Chapter 5 - ammunition/class V material	5.8	26
Chapter 6 - foreign ammunition (NATO)	5.9	26
Chapters, sections, appendixes, index	5.3.1	12
Checklist	5.5.2.5.3.2	19
Component	3.9	5
Components of end item (COEI)	3.10	6
Components of end item (COEI) and basic issue items (BII) lists appendix	5.10.2	26
Conformance to standard	4.1.1	9
Contents	4.2	9
Contracting activity	3.11	6
Cover sheet	5.5.2.5.3.1	18
Decals and instruction plates	5.5.3.5	23
DEFINITIONS	3.	4
DETAILED REQUIREMENTS	5.	11
Development of text and graphics	4.2.1	9
Disassemble	3.12	6
During operation checks	3.13	6
Emergency procedures	5.5.4.3	24
End item	3.14	6
Equipment	3.15	6
Equipment, complex	3.16	6
Equipment illustration	5.5.2.2	14
Expendable and durable items list appendix	5.10.4	29
Fording and swimming	5.5.4.2	24
Front matter	5.3.2	12
Fully mission capable (FMC)	3.17	6
General	5.1	11
GENERAL REQUIREMENTS	4	9

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INDEX

	PARAGRAPH	PAGE
General style requirements	4.1	9
Glossary	4.2.9	10
Glossary	5.10.9	31
Government documents	2.1	2
Hardtime scheduled maintenance	3.18	6
Illustrations	4.2.6	10
Illustrations	5.10.6.1	31
Illustrations	5.5.2.4.1	16
Illustrations	5.5.2.5.1	18
Initial adjustments, before use, daily checks, and self-test (as applicable)	5.5.3.3	21
Intended use	6.1	32
Introduction to PMCS table	5.5.2.1	13
Introductory information	5.6.2.2	25
Issue of DODISS	6.2	32
Jamming and electronic countermeasures (ECM) procedures	5.5.4.5	24
Listing	5.10.6.2	31
Lubrication instructions appendix	5.10.7	31
Malfunction index	5.6.2.3	25
Mandatory replacement parts	5.5.2.3.1	16
Measurements and tolerances	4.2.7	10
Method of presentation	5.5.3.4.1	22
Monthly operation checks	3.19	6
Nomenclature and references to equipment	4.2.3	10
Not fully mission capable (NFMC)	3.21	6
Note	3.20	6
NOTES	6.	32
Nuclear, biological, and chemical (NBC) decontamination procedures	5.5.4.4	24
On-condition scheduled maintenance	3.22	7
On-vehicle equipment loading plan appendix	5.10.6	30
Operating auxiliary equipment	5.5.3.6	23
Operating instructions	5.2.3	11
Operating procedure	5.5.3.4	22
Operating procedures in multilevel maintenance TMs	5.1.1	11
Operation under unusual conditions	4.2.2	10
Operator maintenance	3.23	7
Order of precedence	2.2	3
Other appendixes	5.10.8	31
Other Government documents and publications	2.1.2	3
Packaging and preparation for delivery	5.12	31
Part number (P/N)	3.24	7
PMCS table	5.5.2.3	14
Precombat checklist (PCL)	3.25	7
Precombat checklist (PCL)	5.5.2.5	18
Preparation for movement	5.5.3.7	23
Preparation of TMs	5.2	11
Preventive maintenance checklist (PMC)	5.5.2.4	16
Preventive maintenance (scheduled maintenance)	3.26	7
Purpose	1.2	1
Quality assurance requirements	5.11	31

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Check the source to verify that this is the current version before use.

INDEX

	PARAGRAPH	PAGE
Reference to parts of the equipment	4.2.4	10
References appendix	5.10.1	26
Replace	3.27	7
Safety devices and interlocks	4.2.8	10
Scope	1.1	1
SCOPE	1.	1
Section I - description and use of operator's controls and indicators	5.5.1	12
Section I - introduction	5.10.2.1	27
Section I - introduction	5.10.3.1	29
Section I - lubrication instructions	5.6.1	24
Section II - additional authorized items list	5.10.3.2	29
Section II - components of end item	5.10.2.2	28
Section II - preventive maintenance checks and services (PMCS) and mandatory replacement parts	5.5.2	13
Section II - troubleshooting procedures	5.6.2	25
Section III - basic issue items	5.10.2.3	28
Section III - operation under usual conditions	5.5.3	20
Section III - operator maintenance procedures	5.6.3	26
Section IV - operation under unusual conditions	5.5.4	23
Section V - functional checks	5.5.5	24
Security classification markings	4.1.3	9
Selective application and tailoring	1.4	1
Service	3.28	7
Shelter requirements	5.5.3.1.2	21
Site and shelter requirements	5.5.3.1	20
Siting	5.5.3.1.1	20
Spare part	3.29	7
Specifications, standards, and handbooks	2.1.1	2
Stowage and decal/data plate guide appendix	5.10.5	30
Stowage location	5.10.2.4	28
Style and format requirements	5.5.2.4.2	16
Style and format requirements	5.5.2.5.2	18
Subject term (key word) listing	6.6	33
Supersession data	6.5	32
System	3.30	7
System integration (SI) manual	3.31	7
Tailoring guidance	6.3	32
Technical content	5.5.2.4.3	17
Technical content	5.5.2.5.3	18
Technical content preparation	4.1.2	9
Technical manuals (TMs)	3.32	7
Text content	5.2.1	11
Troubleshooting procedures	5.2.2	11
Unit	3.33	8
Unusual environment/weather	5.5.4.1	24
Usage note and reporting errors statement	5.5.2.4.5	17
Use of color	4.1.4	9
Warning	3.34	8
Warnings and cautions	5.5.2.4.4	17
Weekly operation checks	3.35	8

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3. DOCUMENT TITLE
Operator and Preventive Maintenance Instructions

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME USAMC Materiel Readiness Support Activity	b. TELEPHONE *(Include Area Code)* (1) Commercial: (606) 293-3469 (2) AUTOVON: 745-3469
c. ADDRESS *(Include Zip Code)* ATTN: AMXMD-EP (TMSS) Lexington, KY 40511-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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